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ORIGINAL RESEARCH

Full-state discrete-time model-based stability analysis and parameter design of dual active bridge converter in energy storage system

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Abstract

The dual active bridge (DAB) converter plays a crucial role in energy storage system application of DC microgrid. In such a cascade system, maintaining its stability is imperative for reliable operation of the DAB converter under constant current, constant power, and constant voltage charging mode. Firstly, the accurate discrete-time models considering parasitic resistances are applied for three charging modes to reveal the system dynamic characteristics. The discrete-time small-signal models for different operation modes are developed further which gives guidance in the controller's parameters design. Secondly, the comprehensive and systematic analysis of all the possible unstable phenomena for three charging modes are conducted according to the discrete-time model. Finally, the underlying mechanisms are revealed by analyzing the relationships among the state variables, the Floquet multipliers, and system parameters. It is demonstrated that the inductor and output capacitor of the DAB converter, the DC bus voltage, the equivalent series resistance of the battery, and the proportional constant can influence the system stability under all charging modes greatly. Hence, the stability-oriented parameter design process is given which can provide the guidance in practical. The theoretical analysis under these three types operating modes is verified by simulations and experimental results.

1 | INTRODUCTION

The DC microgrid has attracted worldwide attention due to the development of distributed renewable sources, energy storage system (ESS) and the usage of modern DC loads [1–3]. It has broad application advantages including higher efficiency [4], no reactive power requirement, and elimination of the need for AC–DC or DC–AC conversion stages. In the meantime, ESS can exchange energy with the DC bus based on situations whether the source energy is in deficient or excess state [5–7]. The effective operation of microgrid depends on the optimum management of ESS, which remains as a key challenge [8]. A typical structure of DC microgrid system is shown in Figure 1. Dual active bridge (DAB) converter is suitable for DC microgrid system for its salient merits of high-frequency galvanic isolation, bidirectional power transmission and wide voltage conversion gain range [9, 10]. Therefore, a DAB converter is used to connect the DC bus and ESS in a DC microgrid here.

Stable operation of DC microgrid integrated with ESS is a major challenge where some undesired oscillations and other unexpected unstable phenomena may occur due to the oscillation of the DC-bus voltage [11], improper selection of parameters and control strategies [8]. Moreover, the stability margins are different in different operating modes [12]. In the charging process of ESS, the DAB converter is controlled to realize different modes, such as constant current (CC) mode, constant power (CP) mode, and constant voltage (CV) mode. In order to extend the lifetime of a battery, it is necessary for the DAB converter to provide accurate charge current and voltage through stable operations [13]. If there are unstable phenomena, the corresponding fast-scale subharmonic oscillation and the slow-scale low-frequency oscillation will lead to the undesired current and voltage levels, which can increase the

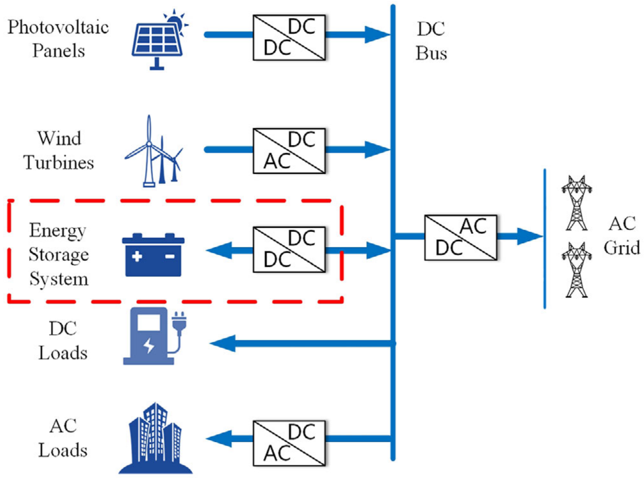

FIGURE 1 Structure of DC microgrid system with ESS

devices stress, decrease their reliability and even damage the DAB converter and its interconnected equipment [14]. To investigate the stability of the system, it is essential to apply proper modelling methods and analysis tools to reveal the complex dynamic behaviours of the system and facilitate the design of high-performance controllers [15].

The previous work of modelling the DAB converter is briefly summarized as follows. Conventional state-space averaging method [16] is widely used to model the converters. However, because the inductor current of DAB converter is pure AC, the small ripple hypothesis no longer holds [14]. Therefore, the generalized averaging method [17] is proposed to investigate the small-signal characteristics of DAB converter [18]. In this model, only the DC term and first order terms of the inductor current are considered, which may result in relatively large error especially when there are large harmonic distortions. There are also the reduced-order impedance models of the DAB converters in [19–22] which neglect the dynamics of the AC conversion stage. As a result, these models are suffering in low accuracy in high frequencies [11]. Thus, they cannot predict slow- or fast-scale stability issues. The methods mentioned above ignore inductor current dynamics, but which can be considered by discrete-time models [23].

The discrete-time models can explicitly describe state trajectories in all sub-intervals of converter operation, which means they are capable of providing exact solutions for AC state variables. Discrete-time models have been shown to be very reliable and accurate, which have been successfully used to study the nonlinear dynamics of switching power converters such as DAB converters. Discrete-time models can also predict both the slow- and fast-scale dynamics of the system. The discrete-time small-signal model of the DAB converter is developed in [24]. Besides, the bilinear discrete-time model of the DAB converter is proposed in [25] to simplify the complex exponential matrices iteration, which is also used to study the nonlinear behaviour of the converter. By using approximate discrete-time models considering various practical parasitic, a discrete-time framework is proposed to analyze the stability of DAB converter at high frequency in [26]. Ref. [14] presents a decomposed discrete-time model and conclude the multi-scale oscillations analysis of the DAB converter. However, literatures above do not employ the discrete-time model to depict the system characteristics of the DAB converter under different charging modes integrated with ESS in DC microgrid. Ref. [12] presents the stability of a two-stage single-phased grid-connected bidirectional EV charger using simplified small-signal impedance models. However, it doesn't consider the stability of DAB converter under CP charging mode. Ref. [11] proposes two impedance compensation methods to enhance the stability of the DAB converter-based ESS. But its model does not involve the parasitic resistance which may significantly affects the power transmission characteristics of DAB converter under different operating conditions [27].

However, throughout the existing literatures, there are few reports using the accurate discrete-time model to analyze three charging modes. Furthermore, the discrete-time small-signal models for CC, CP and CV charging modes are essential to design the PI controller's parameters. Meanwhile, there is little literature on the full-state stability analysis of battery charging based on DAB converter. Therefore, this paper proposes the accurate discrete-time models of the DAB converter integrated with ESS under three charging modes in DC microgrid. By considering the parasitic resistances including the winding resistance and the on-resistance of switching devices, the model can depict the dynamic behaviour of the system precisely. In addition, the discrete-time small-signal models for CC, CP and CV charging modes are established respectively, which provides guidance for engineering application. According to the discrete-time model, the stability of the DAB converter connected with ESS is systematically analyzed under all modes of operation. The relationships among the state variables, Floquet multipliers, and system parameters are obtained through the correlation factor and sensitivity analysis, which can reveal the underlying mechanism of unstable phenomenon. The influence of crucial parameters on the stability of the DAB-based ESS is also investigated. Finally, the parameter spaces of the significant system parameters are obtained for three charging modes. By comparing the stability boundaries among different charging modes, this paper provides reliable guidance for the parameters design and ensure the stable operation of the system.

The rest of this paper is organized as follows. In Section 2, the system description including the circuit configuration and the conventional single-phase-shift control is presented. The discrete-time modelling of the power circuit is analysed in detail. Furthermore, the discrete-time small-signal models of three charging modes are presented which give advices for the design process of PI controller's parameters. In Section 3, the stability analysis of different charging modes is provided respectively. In Section 4, the discussions among three control methods are given based on Section 3. In addition, the optimized design guideline of the DAB converter-based ESS in DC microgrid is proposed. Simulation and experimental results are illustrated to verify the above

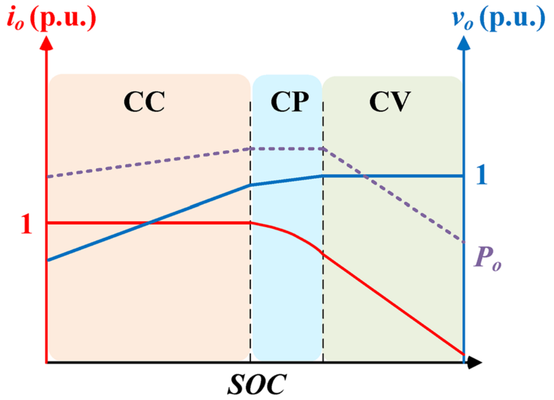

FIGURE 2 CC, CP and CV charging profile of the battery

theoretical analysis in Section 5. Finally, conclusion is provided in Section 6.

2 | SYSTEM DESCRIPTION AND FULL-STATE DISCRETE-TIME MODELING

This section conducts the discrete-time model of the DC microgrid system. At the same time, the design process of PI controller is provided based on discrete-time model. Furthermore, to model and analyze the system more accurately in the following chapters, the influence of factors such as battery internal resistance R_{eq}, on-resistance of switching devices, transformer windings, and line impedance will be considered.

2.1 | System description

In the charging process of the ESS, the DAB converter could be operating in CC charging mode firstly, in this process the battery voltage gradually increases. When the battery voltage increases to a certain range, CP charging mode is introduced in order to reduce the capacity of charging converter [28]. In this process, the battery voltage continues to rise. When the battery voltage reaches the certain charge voltage (for example, 95% of the rated voltage), the DAB converter changes its mode from the CP mode to the CV mode. The charging process is completed when the battery current decreases to a certain value (for example, 0.02C) [13]. All these three scenarios of charging process are considered in the paper which are shown in Figure 2.

The DAB converter studied in this paper will be applied in the DC microgrid integrated with ESS, where the input and the output of the DAB converter are DC bus and energy storage device respectively. Internal resistance equivalent circuit model is utilized to model the battery. The diagram of the system topology and control block is shown in Figure 3. There are two H bridges which consist of eight controllable power switches S_1–S_4 and Q_1–Q_4. A magnetic network which comprises a high frequency transformer with 1:n turn ratios and the external auxiliary inductor can provide not only the galvanic isolation but also the voltage matching. L can be recognized as the instantaneous energy storage device which contains the external inductance and the equivalent transformer leakage inductance. R_t is the sum of the transformer winding resistance and the on-resistance of eight switches. V_1 and V_2 denote the DC bus voltage and the battery voltages respectively. R_{eq} represents the equivalent internal resistances of the battery. C is the output capacitor. i_{ref} is the reference value of output current. P_{ref} is the reference value of output power. V_{ref} is the reference value of output voltage. k_p and k_i are the proportion and integration coefficients of PI controller respectively. φ is the phase shift angle which can determine the direction and magnitude of the power flow.

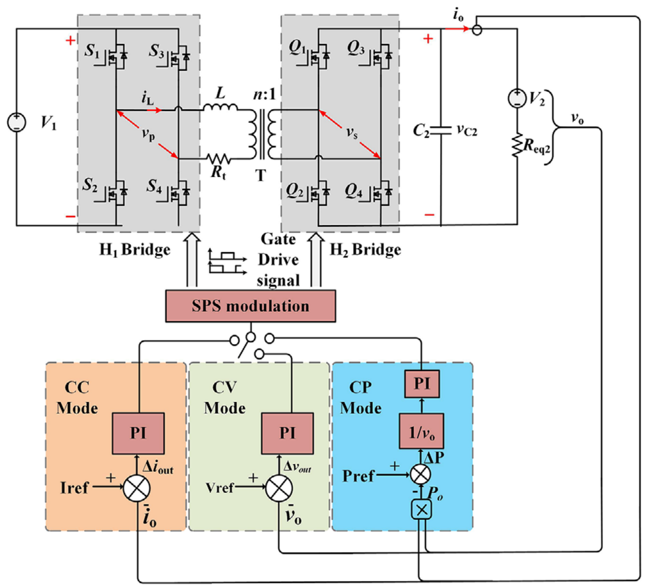

FIGURE 3 System topology and control block diagram

The phase-shift modulation is applied to the controller. There are four typical phase-shift modulation schemes for the DAB converters including the single-phase-shift (SPS), extended-phase-shift (EPS), dual-phase-shift (DPS), and the triple-phase-shift (TPS) modulation. Since the control method in this paper has little to do with the modulation method, the traditional SPS control is selected in order to simplify the system control and model. Here, the forward power flow is from side V_1 to V_2 while the reverse power flow is from side V_2 to V_1 as shown in Figure 3. In addition, the polarities of the phase shift angle in the forward and reverse power flows are positive and negative respectively.

Figure 4 illustrates the operation waveforms in the steady state when power flows in the positive direction. v_p and v_s are the AC voltages of the magnetic network terminals. They are depended on the phase shift angle φ_n in each switching cycle. Therefore, there are four subintervals of the length t_{ni} (i = 1,2,3,4) during one switching cycle in which state variable i_L is symmetric.

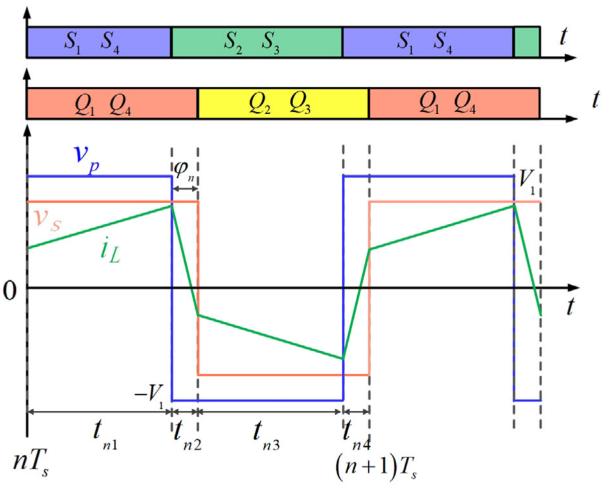

FIGURE 4 Operation waveforms in the steady state

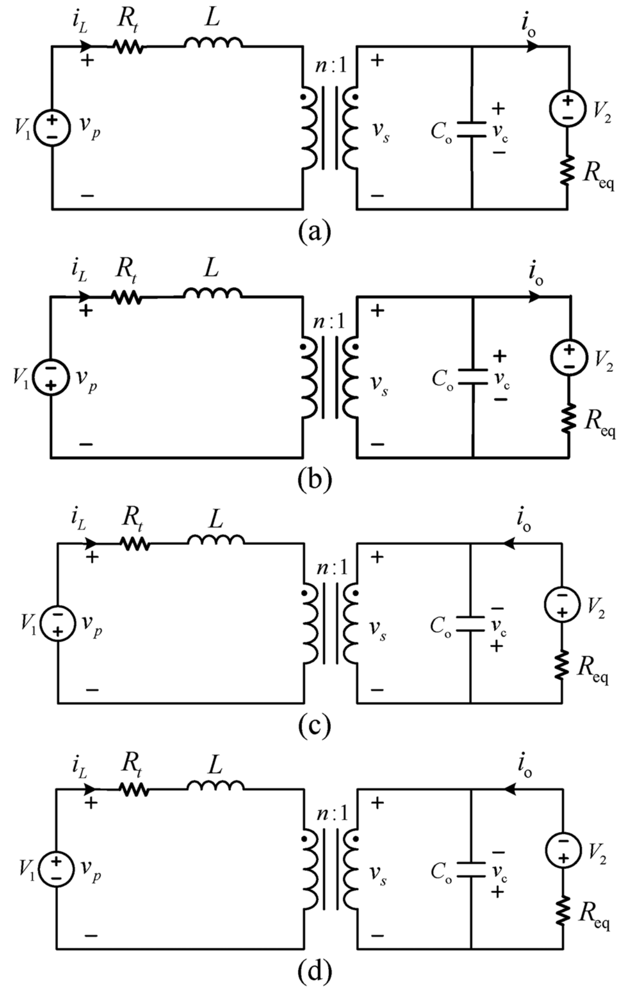

FIGURE 5 Equivalent circuits during subintervals t_{n1}, t_{n2}, t_{n3}, and t_{n4}. (a) Subinterval t_{n1}, (b) subinterval t_{n2}, (c) subinterval t_{n3}, (d) subinterval t_{n4}

2.2 | Discrete-time modelling of power circuit

The DAB converter ranges among the four circuit topologies within a switching cycle as the switches periodically change their configurations. Equivalent circuits during each subinterval t_{n1}, t_{n2}, t_{n3} and t_{n4} are shown in Figure 5.

According to Figure 5, the system state equation of the ith subinterval with the corresponding topology takes the form as:

$$\dot{\mathbf{x}} = \mathbf{A}_i\mathbf{x} + \mathbf{B}_i,\ \ i = 1, 2, 3, 4, \tag{1}$$

where $\mathbf{x}$ is a state vector consisting of the inductor current i_L, and the output capacitor voltage v_C. In addition, the system matrix $\mathbf{A}_i$ and vector $\mathbf{B}_i$ for each topology can be expressed as follows.

$$\mathbf{A}_1 = \mathbf{A}_2 = \begin{bmatrix} -\frac{R_t}{L} & -\frac{n}{L} \\ \frac{n}{C} & -\frac{1}{CR_{eq}} \end{bmatrix},\ \mathbf{A}_3 = \mathbf{A}_4 = \begin{bmatrix} -\frac{R_t}{L} & \frac{n}{L} \\ -\frac{n}{C} & -\frac{1}{CR_{eq}} \end{bmatrix}. \tag{2}$$

$$\mathbf{B}_1 = \mathbf{B}_4 = \begin{bmatrix} \frac{v_1}{L} \\ \frac{v_2}{R_{eq}C} \end{bmatrix},\ \mathbf{B}_2 = \mathbf{B}_3 = \begin{bmatrix} -\frac{v_1}{L} \\ \frac{v_2}{R_{eq}C} \end{bmatrix}. \tag{3}$$

The discrete iterative processes in formula (4) can be obtained from the time integration of four state equations in formula (1) within their corresponding working periods. In a switching cycle, the state vector at the end of each switching subinterval can be used as the initial value for the next subinterval during iteration. The iterative function f_{tni} of each switch subinterval accurately describes the motion trajectory of the state vector $\mathbf{x}$ in the state space by time. Figure 6 shows the iterative relationship of the state variables in one switching cycle.

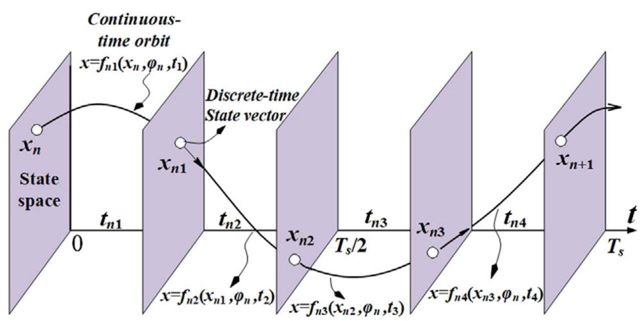

FIGURE 6 State variables iteration relationship

$$\begin{aligned} \mathrm{x}_{n1} &= f_{t_{n1}}(\mathrm{x}_n, t_{n1}) = e^{\mathbf{A}_1 t_{n1}}\mathrm{x}_n + \psi_1, \\ \mathrm{x}_{n2} &= f_{t_{n2}}(\mathrm{x}_{n1}, t_{n2}) = e^{\mathbf{A}_2 t_{n2}}\mathrm{x}_{n1} + \psi_2, \\ \mathrm{x}_{n3} &= f_{t_{n3}}(\mathrm{x}_{n2}, t_{n3}) = e^{\mathbf{A}_3 t_{n3}}\mathrm{x}_{n2} + \psi_3, \\ \mathrm{x}_{n4} &= f_{t_{n4}}(\mathrm{x}_{n3}, t_{n4}) = e^{\mathbf{A}_4 t_{n4}}\mathrm{x}_{n3} + \psi_4, \end{aligned} \tag{4}$$

where

$$t_{n1} = t_{n3} = T_s/2 - (\varphi_n T_s) / (2\pi),$$
$$t_{n2} = t_{n4} = (\varphi_n T_s) / (2\pi),$$
$$\psi_1 = \int_0^{t_1} e^{\mathbf{A}_1 t_{n1}} B_1 \mathrm{d}t = \mathbf{A}_1{}^{-1} \left(e^{A_1 t_{n1}} - \mathrm{I}\right) \mathrm{B}_1,$$
$$\psi_2 = \int_0^{t_2} e^{\mathbf{A}_2 t_{n2}} B_2 \mathrm{d}t = \mathbf{A}_2{}^{-1} \left(e^{A_2 t_{n2}} - \mathrm{I}\right) \mathrm{B}_2, \tag{5}$$
$$\psi_3 = \int_0^{t_3} e^{\mathbf{A}_3 t_{n3}} B_3 \mathrm{d}t = \mathbf{A}_3{}^{-1} \left(e^{A_3 t_{n3}} - \mathrm{I}\right) \mathrm{B}_3,$$
$$\psi_4 = \int_0^{t_4} e^{\mathbf{A}_4 t_{n4}} B_4 \mathrm{d}t = \mathbf{A}_4{}^{-1} \left(e^{A_4 t_{n4}} - \mathrm{I}\right) \mathrm{B}_4.$$

The discrete-time iterative relationship between the state vectors $\mathbf{x}_n$ and $\mathbf{x}_{n+1}$ of the two switching cycles is expressed as:

$$\mathrm{x}_{n+1} = f(\mathrm{x}_n, \varphi_n) = \mathbf{\Phi}\mathrm{x}_n + \psi(\varphi_n), \tag{6}$$

where

$$\begin{aligned}\mathbf{\Phi} &= e^{\mathbf{A}_4 t_{n4}} e^{\mathbf{A}_3 t_{n3}} e^{\mathbf{A}_2 t_{n2}} e^{\mathbf{A}_1 t_{n1}},\\ \psi(\varphi_n) &= e^{\mathbf{A}_4 t_{n4}} \cdot e^{\mathbf{A}_3 t_{n3}} \cdot e^{\mathbf{A}_2 t_{n2}} \psi_1\\ &\quad + e^{\mathbf{A}_4 t_{n4}} \cdot e^{\mathbf{A}_3 t_{n3}} \psi_2\\ &\quad + e^{\mathbf{A}_4 t_{n4}} \psi_3 + \psi_4.\end{aligned} \tag{7}$$

In the above equations, $\mathbf{\Phi x}_n$ represents the free movement of the initial state of the converter in one switching cycle. $\mathbf{\Phi}$ is determined by the structure and parameters of the converter, thus a constant matrix. ψ represents the forced movement of the state variable under the excitation of the input and output voltage sources, so it is not a constant matrix but controlled by the phase shift angle in each switching cycle.

2.3 | Discrete-time small-signal modelling for three charging modes

The DAB converter has multiple operating modes such as CC, CP and CV charging mode. To design the PI controller, the discrete-time small-signal model of each case will be illustrated in this part. The inherent control delay is inevitable in the digital controller. Therefore, the discrete-time model of the PI controller considering the one-step control delay can be obtained as:

$$\begin{aligned}\varphi(n+1) &= \varphi(n) + \Delta(\varphi(n)),\\ \Delta(\varphi(n)) &= K_p \Delta e(n) + K_I e(n)\\ &= K_p(e(n) - e(n-1)) + K_I e(n)\\ &= k_1 e(n) + k_2 e(n-1).\end{aligned} \tag{8}$$

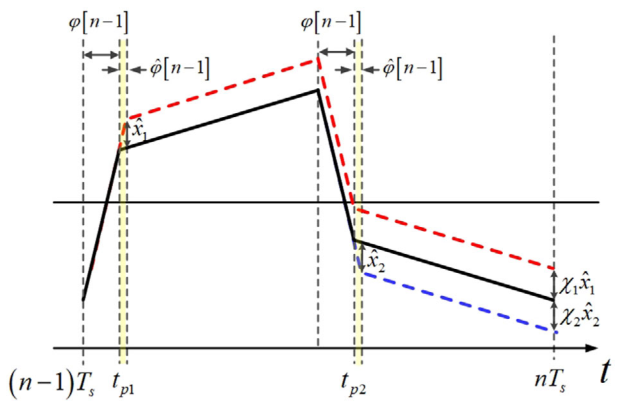

FIGURE 7 Schematic diagram of the influence of phase shift angle small signal disturbance on state variables

where K_p and K_I are the proportional and integral constant respectively. $k_1 = K_p + K_I$, $k_2 = -K_p$, $K_I = K_p T_s / T_I$. T_s and T_I are the switching period and the integration time constant. The error values $e(n)$ are formed differently in each mode which can be obtained as:

$$\begin{cases} e(n) = i_{\mathrm{ref}n} - i_{on} = i_{\mathrm{ref}n} - \left(\begin{bmatrix} 0 & \frac{1}{R_{eq}} \end{bmatrix} \mathrm{x}_n - \frac{V_2}{R_{eq}}\right) & CC\ Mode,\\ e(n) = \dfrac{P_{\mathrm{ref}n} - P_{on}}{v_{cn}} = \dfrac{P_{\mathrm{ref}n} - v_{cn} \times i_{on}}{v_{cn}} & \\ \quad = \dfrac{P_{\mathrm{ref}n} - [0 \quad 1]\,\mathrm{x}_n \times \left(\begin{bmatrix} 0 & \frac{1}{R_{eq}} \end{bmatrix} \mathrm{x}_n - \frac{V_2}{R_{eq}}\right)}{v_{cn}} & CP\ Mode,\\ e(n) = v_{\mathrm{ref}n} - v_{cn} = v_{\mathrm{ref}n} - [0 \quad 1]\,\mathrm{x}_n & CV\ Mode. \end{cases} \tag{9}$$

On the basis of the discrete-time model of DAB converter described in the previous section, small-signal disturbances can be applied to the state variables, then the small-signal model of DAB converter can be obtained and provide guidance for the design of PI controller [29]. The next goal is to establish a small-signal model in the form:

$$\hat{\mathbf{x}}[n] = \mathbf{\Phi}\hat{\mathbf{x}}[n-1] + \mathbf{G}\hat{\varphi}[n-1], \tag{10}$$

where $\hat{\varphi}[n-1]$represents the small-signal disturbances imposed on the phase shift angle $\varphi[n-1]$ in n−1th switching cycle.

After the small-signal disturbances are applied to the phase shift angle φ of the n−1th switching cycle of the DAB converter, the schematic diagram of its influence on the state variables is shown in Figure 7. After applying small signal perturbation to the phase shift angle, small signal perturbations $\hat{x}_1$ and $\hat{x}_2$ of state variables will be generated in the $\hat{\varphi}[n-1]/2\pi f_s$ time period near t_{p1} and t_{p2}. It is considered that $\mathbf{X}_{\mathrm{p1}}$ and $\mathbf{X}_{\mathrm{p2}}$ represent the steady-state values of the state variables at t_{p1} and t_{p2} which are shown in (11). The exponential matrix is expressed by first-order Taylor expansion.

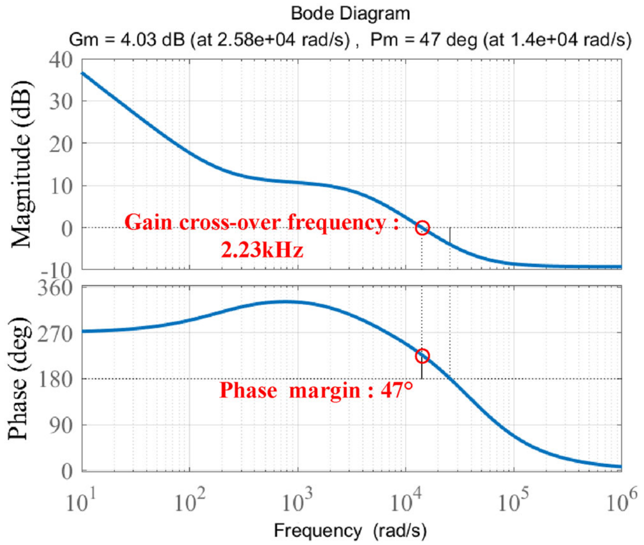

FIGURE 8 Bode diagram of the open-loop transfer function for CC charging mode

$$\begin{aligned}\hat{\mathbf{x}}_1 &= e^{A_4 \hat{t}} X_{p1} + A_4^{-1}(e^{A_4 \hat{t}} - I)B_4 - \left(e^{A_1 \hat{t}} X_{p1} + A_1^{-1}(e^{A_1 \hat{t}} - I)B_1\right) \\ &= (A_4 - A_1)\, X_{p1} \frac{\hat{\varphi}}{2\pi f_s}, \\ \hat{\mathbf{x}}_2 &= e^{A_2 \hat{t}} X_{p2} + A_2^{-1}(e^{A_2 \hat{t}} - I)B_2 - \left(e^{A_3 \hat{t}} X_{p2} + A_3^{-1}(e^{A_3 \hat{t}} - I)B_3\right) \\ &= (A_2 - A_3)\, X_{p2} \frac{\hat{\varphi}}{2\pi f_s}.\end{aligned} \tag{11}$$

The state variable perturbation $\hat{x}_1$ through t_{p1} to nT_s will become $\chi_1 \hat{x}_1$. Similarly, the state variable perturbation $\hat{x}_2$ through t_{p2} to nT_s will become $\chi_2 \hat{x}_2$. Since the state transition matrix $e^{\mathbf{A}_i t_{ni}}$ represents the free transfer process of the state vector in the state space of the DAB converter with time changing, while the input voltage source will not have any influence on the transfer process. The expression of $\chi_1 \hat{x}_1$ and $\chi_2 \hat{x}_2$ are:

$$\begin{aligned}\chi_1 \hat{\mathbf{x}}_1 &= e^{\mathbf{A}_3 t_{n3}} e^{\mathbf{A}_2 t_{n2}} e^{\mathbf{A}_1 t_{n1}} \hat{\mathbf{x}}_1, \\ \chi_2 \hat{\mathbf{x}}_2 &= e^{\mathbf{A}_3 t_{n3}} \hat{\mathbf{x}}_2.\end{aligned} \tag{12}$$

The sum of $\chi_1 \hat{x}_1$ and $\chi_2 \hat{x}_2$ are the small signal disturbances of the state variable caused by the small signal disturbance of the phase shift angle. Therefore, the relationship $\mathbf{G}$ between $\hat{\varphi}[n-1]$ and $\hat{x}[n]$ can be expressed as:

$$\begin{aligned}\mathbf{G} &= \left(\chi_1 \hat{\mathbf{x}}_1 + \chi_2 \hat{\mathbf{x}}_2\right) / \hat{\varphi} \\ &= \chi_1 \left(\mathbf{A}_4 - \mathbf{A}_1\right) \mathbf{X}_{p1} \frac{1}{2\pi f_s} + \chi_2 \left(\mathbf{A}_2 - \mathbf{A}_3\right) \mathbf{X}_{p2} \frac{1}{2\pi f_s}.\end{aligned} \tag{13}$$

Similarly, since the state transition matrix $e^{\mathbf{A}_i t_{ni}}$ represents the transition process of the state vector in the state space over time, the relationship $\mathbf{\Phi}$ between $\hat{x}[n]$ and $\hat{x}[n-1]$ can be expressed as:

$$\mathbf{\Phi} = e^{\mathbf{A}_4 t_{n4}} e^{\mathbf{A}_3 t_{n3}} e^{\mathbf{A}_2 t_{n2}} e^{\mathbf{A}_1 t_{n1}}. \tag{14}$$

By appropriately modifying formula (10), the discrete domain transfer function from the phase shift angle to the state variable vector of the DAB converter can be obtained as:

$$G_{x\varphi}(z) = \frac{\hat{\mathbf{x}}}{\hat{\varphi}} = \left(z\mathbf{I} - \mathbf{\Phi}\right)^{-1} \mathbf{G}. \tag{15}$$

For different charging modes, the discrete domain transfer functions from control to output are various. The control block and small-signal transfer function is derived for each case as follows.

2.3.1 | CC charging mode

During the CC charging mode, the control objective of the DAB converter is to regulate the battery port DC current. The current regulator control block diagram is shown in Figure 3. A proportional-integral (PI) controller is adopted to improve the dynamic response of the output current control. The measured output current is compared with the i_{ref}, then the phase shift angle is obtained through the PI controller. The relationship between the output current i_o and the state variable can be obtained as:

$$i_o = \frac{v_c - v_2}{R_{eq}}. \tag{16}$$

Combining Equations (15) and (16), the discrete domain transfer function $G_{i_o\varphi}(z)$ from the phase shift angle to the output current can be obtained as shown in Equation (17), where $[0 \quad \frac{1}{R_{\mathrm{eq}}}]$ represents the conversion relationship between the output current and the state variables. The Bode diagram of the open-loop transfer function of the system after PI correction is shown in Figure 8, which is the product of (17) and (20). The parameters are the same as Table 1. The bandwidth is 2.23 kHz, which is more than 1/10 of switching frequency. At the same time, the phase margin is 47° which meets the requirement of engineering application.

$$G_{i_o\varphi}(z) = \frac{\hat{i}_o}{\hat{\varphi}} = \begin{bmatrix} 0 & \frac{1}{R_{\mathrm{eq}}} \end{bmatrix} \left(z\mathbf{I} - \mathbf{\Phi}\right)^{-1} \mathbf{G}. \tag{17}$$

2.3.2 | CP charging mode

During the CP charging mode, the control objective of the DAB converter is to regulate the output power. To directly control the power from the DAB converter to the battery rapidly and accurately, a direct power control scheme (DPC) with single-phase-shift control is proposed for the DAB DC–DC converter. In the executing procedure of the proposed DPC method, two quantities including the output voltage and

TABLE 1 Specifications of the DAB converter prototype

Symbol	Quantity	Values
V_1	DC bus voltage	48 V
L	Sum of the external and leakage inductance	35.49 μH
V_2	Battery voltage in CC mode	44 V
V_2	Battery voltage in CP mode	44 V
V_2	Battery voltage in CV mode	46 V
R_t	Sum of the winding resistance and on-resistance of switching devices	0.15 Ω
C	Output capacitor	500 μF
R_{eq}	Equivalent series resistance of the battery	0.5 Ω
n	Turns ratio of the transformer	1
f_s	Switch frequency	20 kHz
i_{ref}	Reference output current (CC)	3 A
P_{ref}	Reference output power (CP)	120 W
V_{ref}	Reference output voltage (CV)	48 V
K_p	Proportional constant (CC)	0.4
K_p	Proportional constant (CP)	0.4
K_p	Proportional constant (CV)	0.9
K_I	Integral constant	200

the output current should be measured and sampled. The output power regulator control block diagram is shown in Figure 3.

The output power P_o is calculated by multiplying the output current i_o with the output voltage v_{C2}. Because the ΔP is divided by the v_c to obtain the output current reference value, the PI controller is designed based on the relationship between the output current and the phase shift angle. The division operation ensures the constant output power when the voltage fluctuates. The feedback phase shift angle is obtained through the PI controller. Therefore, the small-signal transfer function of CP mode is the same as CC mode. However, due to the difference of parameters, the steady-state operating point is different, so the value of the transfer function is also different. The Bode diagram of the open-loop transfer function of the system after PI correction is shown in Figure 9. The parameters are the same as Table 1. The bandwidth is 2.32 kHz, which is more than 1/10 of switching frequency. At the same time, the phase margin is 44.4° which meets the requirement of engineering application.

2.3.3 | CV charging mode

The control objective of the DAB converter of CV mode is to regulate the voltage at its output port where the battery is connected. The voltage regulator control block diagram is shown in Figure 3. The measured output voltage of the DAB converter is compared with the V_{ref}, then the phase shift angle is obtained through the PI controller. The PI controller is designed based on the relationship between the output voltage v_o and the phase

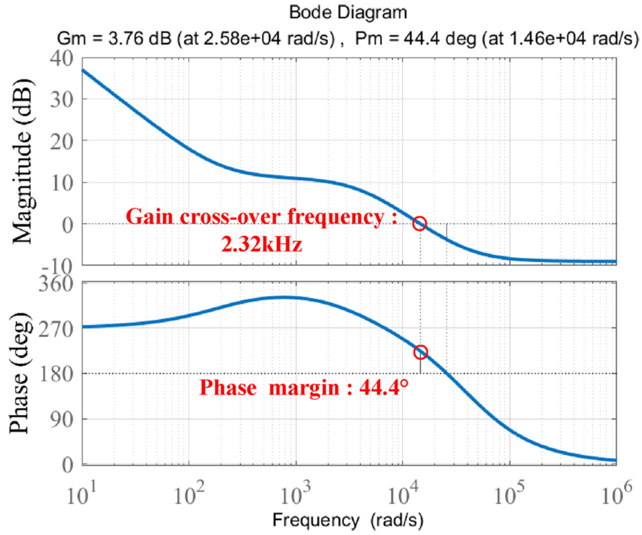

FIGURE 9 Bode diagram of the open-loop transfer function for CP charging mode

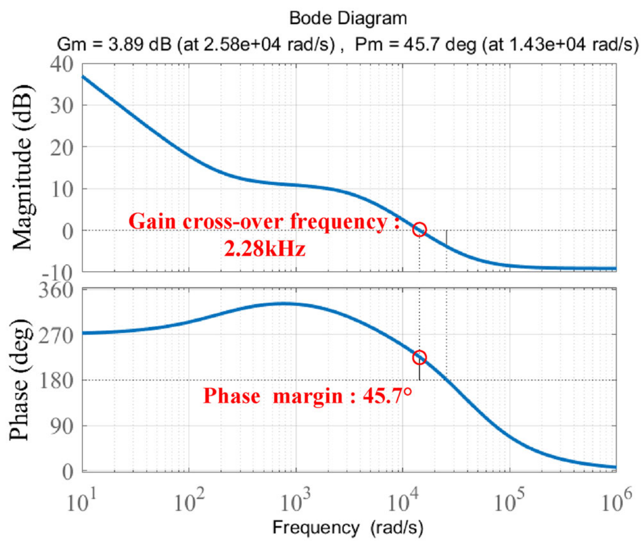

FIGURE 10 Bode diagram of the open-loop transfer function for CV charging mode

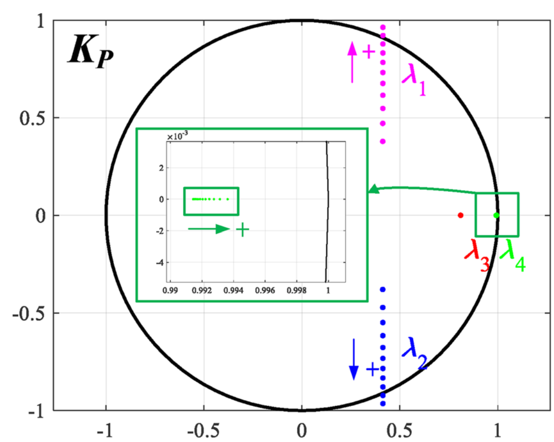

FIGURE 11 The root loci of the Floquet multipliers when K_p increases

shift angle. According to the functional block diagram of the DAB converter system, the relationship between the output voltage and the state variable can be obtained as:

$$v_O = v_c = [0 \quad 1]x_n. \tag{18}$$

Combining Equations (15) and (18), the discrete domain transfer function $G_{v_o\varphi}(z)$ from the phase shift angle to the output voltage can be obtained as shown in Equation (19), where [0 1] represents the conversion relationship between the output voltage and the state variables.

$$G_{v_o\varphi}(z) = \frac{\hat{v}_o}{\hat{\varphi}} = [0 \quad 1] \left(z\mathbf{I} - \mathbf{\Phi}\right)^{-1} \mathbf{G}. \tag{19}$$

In addition, the discrete domain transfer function of the digital PI controller can be expressed in the form of formula (20). According to the transfer function from control to output in formula and the transfer function of the controller in formula (17) and (19), the open-loop transfer function of the system can be finally obtained, which in turn provides guidance for the design of PI controllers. For CV charging mode, the Bode diagram of the open-loop transfer function of the system after PI correction is shown in Figure 10, which is the product of (19) and (20). The parameters are the same as Table 1. The bandwidth is 2.28 kHz, which is more than 1/10 of switching frequency. At the same time, the phase margin is 45.7°which meets the requirement of engineering application.

$$G_{\mathrm{PI}} = K_p + \frac{K_p K_I T_s z}{z - 1}. \tag{20}$$

2.4 | Discrete-time model of the system

The main circuit model of formula (6) and the controller model of formula (8) are combined to obtain the accurate discrete-time model of the entire system, as shown in formula (21).

$$\begin{cases} i_L(n+1) = [1 \quad 0]x(n+1), \\ v_c(n+1) = [0 \quad 1]x(n+1), \\ \varphi(n+1) = \varphi(n) + \Delta(\varphi(n)). \end{cases} \tag{21}$$

where

$$\begin{aligned} \Delta(\varphi(n)) &= K_p \Delta e(n) + K_I e(n) \\ &= K_p (e(n) - e(n-1)) + K_I e(n) \\ &= k_1 e(n) + k_2 e(n-1). \end{aligned} \tag{22}$$

Since the previous derivation process does not make much mathematical approximation and has considered the battery internal resistance, the on-resistance of the switching device and other parameters, the discrete-time model of the control method is highly accurate. It can effectively predict the dynamic behaviour of the DAB converter with the changing of circuit parameters, and the PI controller parameters designed by this model can make the system have good dynamic performance as well. Furthermore, the stability analysis of the charging process of ESS integrated with DAB converter in DC microgrid can be obtained.

3 | STABILITY ANALYSIS OF THREE CHARGING MODES

The system parameters and control algorithms have great effects on the stability of the whole system. Inappropriate system parameters may lead to the obvious oscillation of inductor current and output voltage which is undesired. Therefore, it is important to figure out how the system changes from its fundamental operation to unstable state with the variation of parameters.

This section details the stability considerations of DAB converter integrated with ESS under CC charging mode, CP charging mode and CV charging mode in DC microgrid. The parameter spaces and the stability boundaries of crucial parameters are given through correlation factor and sensitivity analysis.

3.1 | CC charging mode stability assessment

3.1.1 | Jacobin matrix and Floquet multipliers of the system

Jacobin matrix and Floquet multipliers

The stability analysis of the system can be conducted with the help of Floquet multipliers of the corresponding Jacobian matrix. The Jacobian matrix of the CC charging mode-controlled DAB converter is obtained according to the discrete-time model described in Section 2. The DAB converter system under the one-step delay can be expressed as:

$$\begin{cases} i_{\mathrm{L}(n+1)} = \left[1 \ 0\right] \mathbf{x}_{(n+1)}, \\ v_{\mathrm{C}(n+1)} = \left[0 \ 1\right] \mathbf{x}_{(n+1)}, \\ \varphi_{(n+1)} = \varphi_{(n)} + k_1 \left(i_{\mathrm{ref}n} - \mathrm{I_{ex}}\mathbf{x}_n\right) + k_2 \left(i_{\mathrm{ref}(n-1)} - \mathrm{I_{ex}}\mathbf{x}_{(n-1)}\right), \\ \varphi_{(n+1)} \in \left[0, \pi/2\right]. \end{cases} \tag{23}$$

where

$$\mathrm{I_{ex}} = \left[0 \quad 1/R_{eq}\right]. \tag{24}$$

Because the phase shift angle $\varphi_{(n+1)}$ is associated with the state variables in both the nth and $(n-1)$ th switching cycles, the following formula transformations are utilized to rewrite the discrete-time system.

$$\begin{cases} x_{1n} = i_{L(n-1)} \\ x_{2n} = i_{Ln} \\ y_{1n} = v_{C(n-1)} \\ y_{2n} = v_{Cn} \end{cases} \tag{25}$$

When applied by the above formula transformations, the standard form of the discrete-time model of the DAB converter system is finally expressed as:

$$\begin{cases} x_{1(n+1)} = x_{2n}, \\ x_{2(n+1)} = [1 \;\; 0]\left(\mathbf{\Phi}(\varphi_n)\begin{bmatrix} x_{2n} \\ y_{2n}\end{bmatrix} + \boldsymbol{\psi}(\varphi_n)\right), \\ y_{1(n+1)} = y_{2n}, \\ y_{2(n+1)} = [0 \;\; 1]\left(\mathbf{\Phi}(\varphi_n)\begin{bmatrix} x_{2n} \\ y_{2n}\end{bmatrix} + \boldsymbol{\psi}(\varphi_n)\right), \\ \varphi_{(n+1)} = \varphi_{(n)} + k_1\left(i_{\mathrm{ref}n} - \mathrm{I}_{\mathrm{ex}}\begin{bmatrix} x_{2n} \\ y_{2n}\end{bmatrix}\right) + k_2\left(i_{\mathrm{ref}(n-1)} - \mathrm{I}_{\mathrm{ex}}\begin{bmatrix} x_{1n} \\ y_{1n}\end{bmatrix}\right). \end{cases} \tag{26}$$

The Jacobian matrix of the DAB converter system as shown in (26) can be obtained by taking the partial derivatives of (27) on the fixed point (X ϕ).

$$\mathbf{J} = \begin{bmatrix} \frac{\partial x_{1(n+1)}}{\partial x_{1n}} & \frac{\partial x_{1(n+1)}}{\partial y_{1n}} & \frac{\partial x_{1(n+1)}}{\partial x_{2n}} & \frac{\partial x_{1(n+1)}}{\partial y_{2n}} & \frac{\partial x_{1(n+1)}}{\partial \varphi_n} \\ \frac{\partial y_{1(n+1)}}{\partial x_{1n}} & \frac{\partial y_{1(n+1)}}{\partial y_{1n}} & \frac{\partial y_{1(n+1)}}{\partial x_{2n}} & \frac{\partial y_{1(n+1)}}{\partial y_{2n}} & \frac{\partial y_{1(n+1)}}{\partial \varphi_n} \\ \frac{\partial x_{2(n+1)}}{\partial x_{1n}} & \frac{\partial x_{2(n+1)}}{\partial y_{1n}} & \frac{\partial x_{2(n+1)}}{\partial x_{2n}} & \frac{\partial x_{2(n+1)}}{\partial y_{2n}} & \frac{\partial x_{2(n+1)}}{\partial \varphi_n} \\ \frac{\partial y_{2(n+1)}}{\partial x_{1n}} & \frac{\partial y_{2(n+1)}}{\partial y_{1n}} & \frac{\partial y_{2(n+1)}}{\partial x_{2n}} & \frac{\partial y_{2(n+1)}}{\partial y_{2n}} & \frac{\partial y_{2(n+1)}}{\partial \varphi_n} \\ \frac{\partial \varphi_{(n+1)}}{\partial x_{1n}} & \frac{\partial \varphi_{(n+1)}}{\partial y_{1n}} & \frac{\partial \varphi_{(n+1)}}{\partial x_{2n}} & \frac{\partial \varphi_{(n+1)}}{\partial y_{2n}} & \frac{\partial \varphi_{(n+1)}}{\partial \varphi_n} \end{bmatrix}_{(\mathrm{X},\,\phi)} \tag{27}$$

Finally, the Floquet multipliers λ_1, λ_2, λ_3, and λ_4 can be obtained by calculating (28).

$$\det\left(\lambda\mathbf{I} - \mathbf{J}\right) = 0, \tag{28}$$

where $\mathbf{I}$ is the identity matrix of the appropriate size.

Calculation of the fixed point

According to the periodic characteristics of the state variables in the steady state, the discrete iterative relationship of the state variable in the two switching periods can be obtained as:

$$\mathbf{x}_{n+1} = \mathbf{I}_{\mathrm{HC}}\mathbf{x}_n. \tag{29}$$

where

$$\mathbf{I}_{\mathrm{HC}} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{30}$$

Thus, the steady-state operating point of the system can be expressed as:

$$\mathbf{X} = \left(\mathbf{I}_{\mathrm{HC}} - \mathrm{ABCD}\right)^{-1}\left(\mathrm{D}\left(\mathrm{C}\left(\mathrm{B}\left(\psi_1\right) + \psi_2\right) + \psi_3\right) + \psi_4\right) \tag{31}$$

where

$$\mathrm{A} = e^{\mathrm{A}_1 t_{n1}}, \mathrm{B} = e^{\mathrm{A}_2 t_{n2}}, \mathrm{C} = e^{\mathrm{A}_3 t_{n3}}, \mathrm{D} = e^{\mathrm{A}_4 t_{n4}} \tag{32}$$

Since the steady-state error of the DAB converter with PI controller is zero, the phase shift angle φ can in the steady state can be calculated according to:

$$\mathrm{I}_{\mathrm{ex}}\mathrm{X} - \frac{V_2}{R_{eq2}} = i_{ref}. \tag{33}$$

3.1.2 | Multi-scale oscillations of the DAB converter

The digitally controlled DAB converter may encounter different types of oscillations under certain system parameters and working conditions in the practical applications, thus downgrading the system performance and damaging the converter [14]. The oscillation phenomena can mainly be summarized as the slow-scale low-frequency oscillation (Hopf bifurcation) and the fast-scale subharmonic oscillation (Period doubling bifurcation). In the field of power electronics, the state of oscillation is considered to be an unstable state. According to the Jacobian matrix of the DAB converter, the Floquet multipliers can be easily obtained by (28), which determines the stability of the system and the phenomena of oscillation type.

There are four Floquet multipliers λ_1, λ_2, λ_3, and λ_4 of the DAB converter with the parameters in Table 1. The trajectory of the Floquet multipliers is shown in Figure 11 with the variation of proportional constant K_p. When K_p changes from 0.2 to 0.7, the corresponding magnitude and phase of the Floquet multipliers are shown in Figure 12. The Floquet multipliers λ_3 and λ_4 can be found almost remains settled on the real axis within the unit circle. In particular, the Floquet multiplier λ_4 does not exceed the unit circle. In addition, there is a conjugate pair of Floquet multipliers λ_1 and λ_2 moving from inside of the unit circle to the outside. Accordingly, the DAB converter will transit from the stable state to unstable.

Figure 13 illustrates the stable inductor current of the DAB converter with the parameters in Table 1. It can be seen from the magnitude plot of the Floquet multipliers in Figure 12a that when the proportional constant K_p increases to about 0.64, the conjugate pair of Floquet multipliers lies on the unit circle. In this case, the Hopf bifurcation appears in the DAB converter and the inductor current shows the slow-scale low-frequency

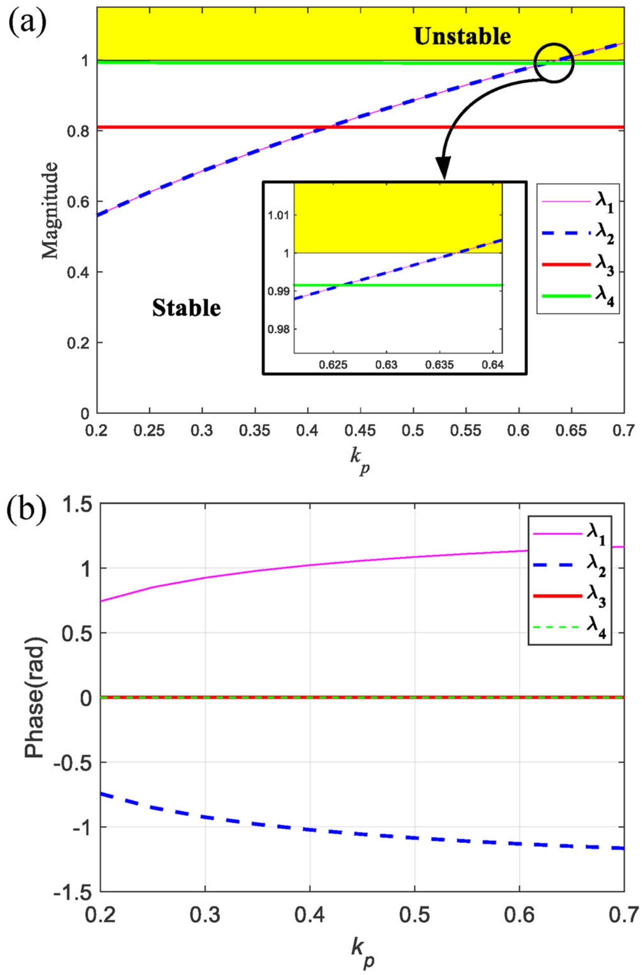

FIGURE 12 The magnitude and phase of the Floquet multipliers. (a) Magnitude, (b) phase

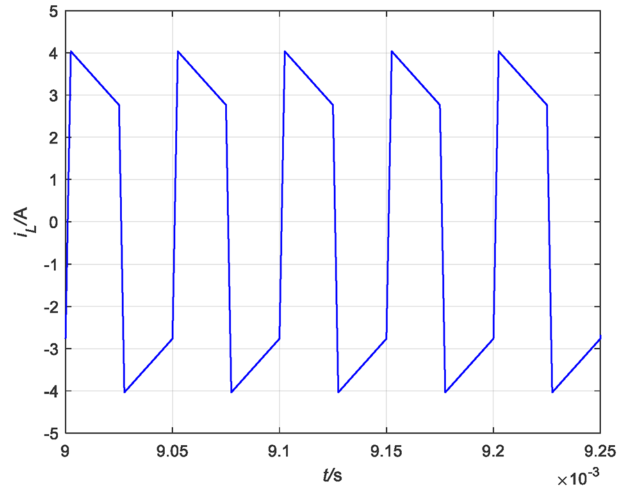

FIGURE 13 Stable state inductor current waveform

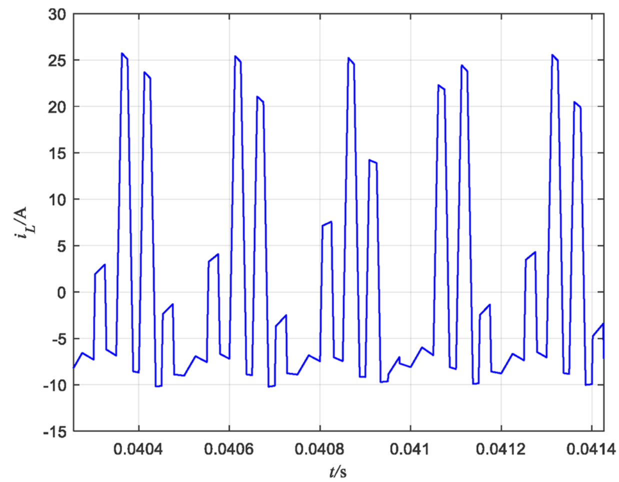

FIGURE 14 Slow-scale low-frequency oscillation of the inductor current

oscillation. The corresponding inductor current is shown in Figure 14. Compared with the steady state, the amplitude of the inductor current is much higher and it is asymmetric within one switching cycle, which may cause the transformer saturation and increase the stress of other equipment. The rest of the paper analyzes the multi-scale oscillations of the DAB converter in detail.

3.1.3 | Stability analysis

Due to the different characteristics of the Floquet multipliers that determine the types of multi-scale oscillations of the DAB converter, this section will analyze Jacobian matrix and Floquet multipliers from different perspectives. According to the correlation factor analysis of the discrete-time DAB converter system proposed in [14], it reveals which state variables have significant impacts on the Floquet multipliers that determine the multi-scale oscillation. Because improper parameters can cause multi-scale oscillations which leads to system instability, this paper conducts the sensitivity analysis and finds out the system parameters that play a key role in the stability of the system. Finally, stable parameter space of the selected key parameters is formed which gives guidance for proper parameter design.

Correlation factor

It is clear that the different forms of the Floquet multipliers and their unique movements have a decisive influence on the dynamic characteristics of the different state variables. In this part, the correlation between different Floquet multipliers and state variables can be analyzed according to the correlation factor of discrete-time systems proposed in [14]. Correlation Factor is a sensitivity-based tool to measure the relative participation of the state variable in the Floquet multipliers. The correlation factors between the state variables and Floquet multipliers of the DAB converter are shown in Table 2 and Figure 15. In addition, the modulus of the correlation factor in Table 2 represents the relevant degree between the state variable and the Floquet multiplier [14].

Obviously, each state variable corresponds to a closely related Floquet multiplier. The fixed Floquet multiplier λ_4 is closely related with the capacitor voltage v_C. In addition, the inductor

TABLE 2 Correlation factor $|\mathbf{P}_{ij}|$

	$\lvert P^{\lambda_i}_{i_L(n)}\rvert$	$\lvert P^{\lambda_i}_{v_C(n)}\rvert$	$\lvert P^{\lambda_i}_{i_L(n+1)}\rvert$	$\lvert P^{\lambda_i}_{v_C(n+1)}\rvert$	$\lvert P^{\lambda_i}_{\varphi(n+1)}\rvert$
λ_1	0.0000	0.7786	0.0004	0.6298	0.5348
λ_2	0.0000	0.7786	0.0004	0.6298	0.5348
λ_3	0.0000	0.0000	0.9993	0.0007	0.0000
λ_4	0.0000	0.7393	0.0000	0.0113	0.2720

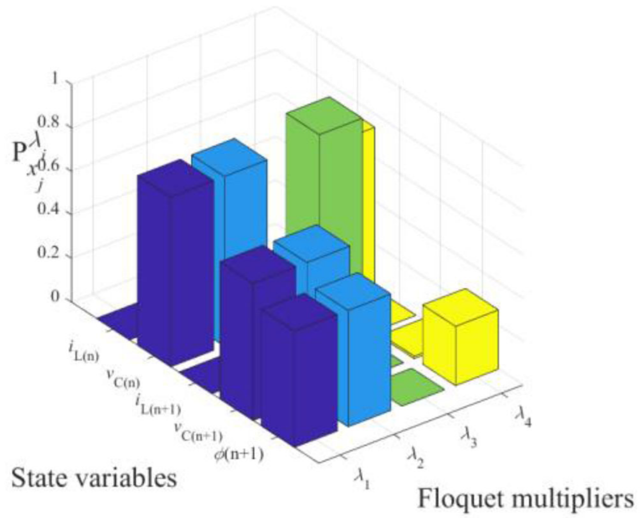

FIGURE 15 The CC charging mode correlation factors between the state variables and Floquet multipliers

current i_L is closely related to the Floquet multiplier λ_3. Besides, the phase shift angle φ and the capacitor voltage v_C is closely related with the Floquet multiplier λ_1 and λ_2, which could result in the low-frequency oscillation.

Sensitivity of the Floquet multipliers

According to the previous analysis, the trajectory of the Floquet multipliers will determine the multi-scale oscillation of the DAB converter and other unstable phenomena. Therefore, in order to prevent the occurrence of instability, it is necessary to study the influence of system parameters on the Floquet multiplier. Thus, the sensitivity analysis of Floquet multipliers to the system parameters of DAB converter is proposed in this part. According to [14], the sensitivity of the Floquet multiplier λ_i to the system parameter K_j is expressed as (34), where $\mathbf{U}$ and $\mathbf{V}$ are the matrices of the right and left eigenvectors of the Jacobian matrix $\mathbf{J}$, respectively.

$$S^{\lambda_i}_{K_j} = \frac{\partial \lambda_i}{\partial K_j} K_j = \frac{\mathbf{V}_i^T \left(\frac{\partial \mathbf{J}}{\partial K_j}\right) \mathbf{U}_i}{\mathbf{V}_i^T \mathbf{U}_i} K_j \tag{34}$$

Table 3 shows the parameter sensitivity of the Floquet multipliers of the DAB converter. The power stage parameters like the inductance L, DC bus voltage V_1, battery voltage V_2, equivalent series resistance R_t, output capacitor C, equivalent series

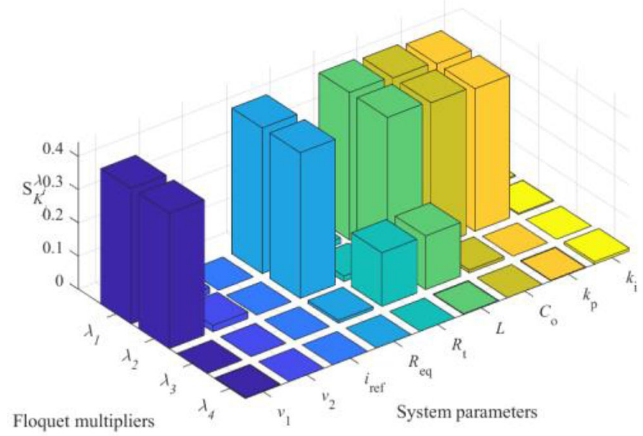

FIGURE 16 The CC charging mode sensitivity of the Floquet multipliers

resistance of the battery R_{eq}, and the digital controller parameters like the proportional constant k_p, integral constant k_I and the output current reference i_ref are selected to conduct the sensitivity analysis.

The sensitivity of the Floquet multipliers is illustrated in Figure 16, where the height of each bar represents the modulus of each element in Table 3. Due to the conjugate pair of Floquet multipliers λ_1, λ_2 have relations with the low-frequency oscillation which lead to the instability of the system, the sensitivity analysis of these two Floquet multipliers is mainly considered. As can be seen from the Figure 16, the conjugate pair of Floquet multipliers λ_1, λ_2 are related to the low-frequency oscillation have very low sensitivity to system parameters like battery voltage V_2, equivalent series resistance R_t, the output current reference i_ref. It indicates that when the voltage of the energy storage system fluctuates, or the i_ref step changes, or the equivalent series resistance R_t varies, the system is not sensitive to the changes of these parameters and can still maintain stable.

The conjugate pair of Floquet multipliers λ_1, λ_2 have high sensitivity to the DC bus voltage V_1. Therefore, the oscillation of V_1 the may influence the stability of system greatly. The signs of the image parts of $S^{\lambda_1}_{V_1}$ and $S^{\lambda_1}_{V_2}$ are the same as those of λ_1 and λ_2, respectively. It indicates that λ_1 and λ_2 will gradually approach the unit circle with the increase of V_1 and the system stability margin will accordingly shrink.

For the parameters of DAB converter, in Table 3, the signs of the image parts of $S^{\lambda_1}_{C}$ and $S^{\lambda_2}_{C}$ are opposite as those of λ_1 and λ_2 respectively, which indicates that λ_1 and λ_2 will gradually move away from the unit circle with the increase of C and the system stability margin will accordingly expand. Therefore, the larger the C, the more stable the system. It is noticeable that the signs of the image parts of $S^{\lambda_1}_{L}$and $S^{\lambda_2}_{L}$are opposite to those of λ_1 and λ_2, which indicates that λ_1 and λ_2 will gradually move away from the unit circle with the increase of L and the system stability margin will accordingly expand.

TABLE 3 Parameter sensitivity of the DAB converter

	λ_1	λ_2	λ_3	λ_4
$S_{V_1}^{\lambda_i}$	$0.0012 + 0.4077i$	$0.0012 - 0.4077i$	$-0.0006 + 0.0000i$	$-0.0019 + 0.0000i$
$S_{V_2}^{\lambda_i}$	$-0.0002 + 0.0195i$	$-0.0002 - 0.0195i$	$0.0005 + 0.0000i$	$-0.0001 + 0.0000i$
$S_{i_{ref}}^{\lambda_i}$	0.0000+0.0000	0.0000+0.0000	0.0000+0.0000	0.0000+0.0000
$S_{R_{eq}}^{\lambda_i}$	$0.0776 - 0.4315i$	$0.0776 + 0.4315i$	$0.0084 + 0.0000i$	$0.0000 + 0.0000i$
$S_{R_t}^{\lambda_i}$	$-0.0046 + 0.0155i$	$-0.0046 - 0.0155i$	$-0.1617 + 0.0000i$	$0.0000 + 0.0000i$
$S_{L}^{\lambda_i}$	$0.0036 - 0.4438i$	$0.0036 + 0.4438i$	$0.1617 + 0.0000i$	$0.0019 + 0.0000i$
$S_{C}^{\lambda_i}$	$0.0777 - 0.4306i$	$0.0777 + 0.4306i$	$0.0085 + 0.0000i$	$-0.0001 + 0.0000i$
$S_{k_p}^{\lambda_i}$	$0.0010 + 0.4295i$	$0.0010 - 0.4295i$	$-0.0000 + 0.0000i$	$-0.0020 + 0.0000i$
$S_{k_I}^{\lambda_i}$	$0.0036 + 0.0004i$	$0.0036 - 0.0004i$	$0.0000 + 0.0000i$	$-0.0073 + 0.0000i$

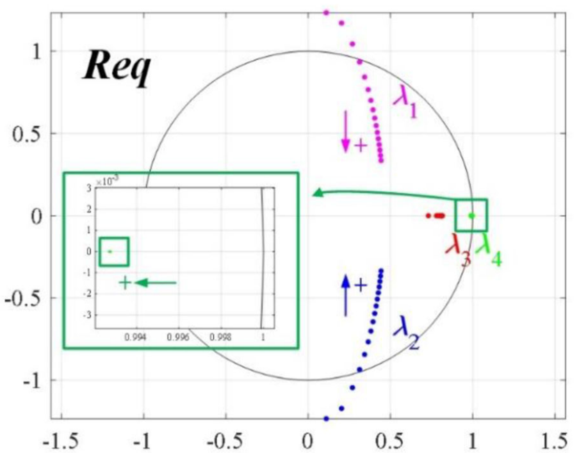

FIGURE 17 The loci of the Floquet multipliers when R_{eq} increases

For equivalent series resistance of the battery R_{eq}, the signs of the image parts of $S_{R_{eq}}^{\lambda_1}$ and $S_{R_{eq}}^{\lambda_2}$ are the opposite as those of λ_1 and λ_2 respectively, which indicates that λ_1 and λ_2 will gradually move away from the unit circle with the increase of R_{eq} and the system stability margin will accordingly expand. It is verified in Figures 17 and 18. Furthermore, with the increase of the battery charge and discharge times, the internal resistance of the battery will slowly increase. Therefore, R_{eq} will not influence the stability of the system, and the CC charging method is not sensitive to the age of the ESS.

For control parameters, it can be obviously discovered that the conjugate pair of Floquet multipliers λ_1, λ_2 have high sensitivity to the proportional constant k_p. However, the Floquet multipliers have quite low sensitivity to the integral constant k_I, so the Floquet multipliers almost remain settled when k_I varies. Therefore, when designing control parameters, it is necessary to accurately design k_p to ensure system stability. It is also noticeable that the signs of the image parts of $S_{k_p}^{\lambda_1}$ and $S_{k_p}^{\lambda_2}$ are the same as those of λ_1 and λ_2, respectively. It indicates that λ_1 and λ_2 will gradually approach the unit circle with the increase of k_p and the system stability margin will accordingly shrink. This conclusion can be verified by Figures 11 and 12.

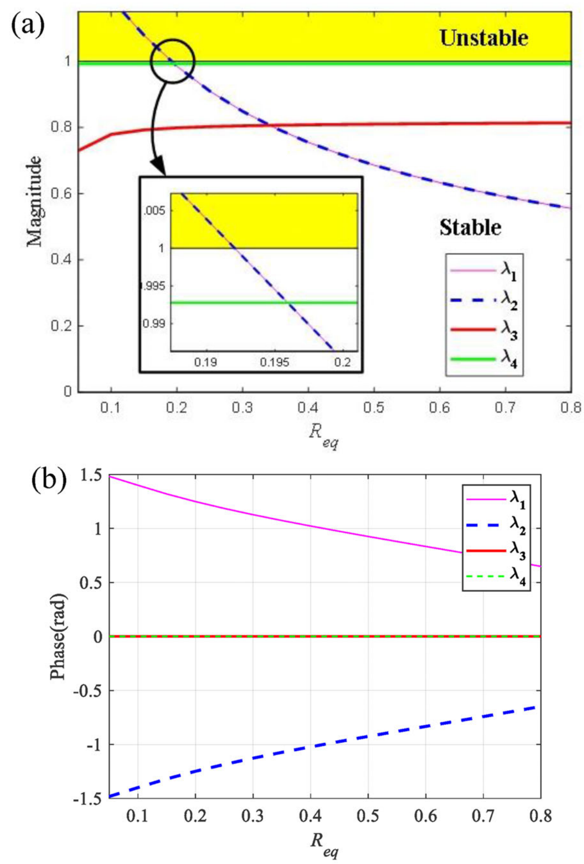

FIGURE 18 The magnitude and phase of the Floquet multipliers. (a) Magnitude, (b) phase

Parameter spaces and stability boundaries

According to the sensitivity analysis of the Floquet multipliers, the DC bus voltage V_1, the equivalent series resistance of

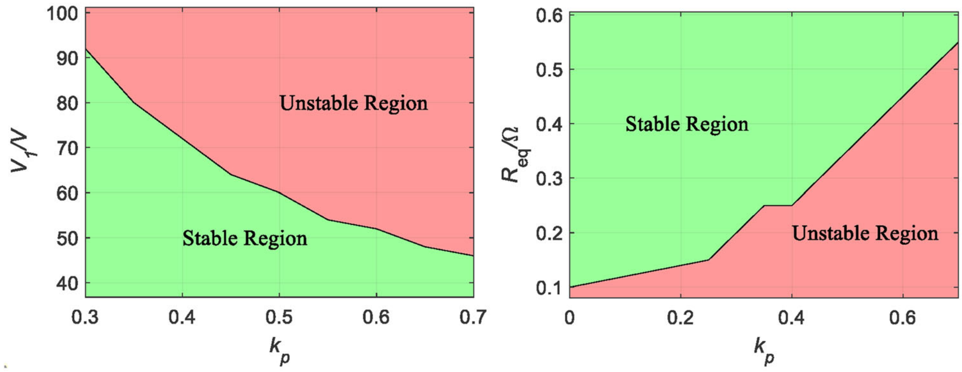

FIGURE 19 The parameter spaces and the stability boundaries of CC charging mode

TABLE 4 Correlation factor $|\mathbf{P}_{ij}|$

	$\lvert P^{\lambda_i}_{i_L(n)}\rvert$	$\lvert P^{\lambda_i}_{v_C(n)}\rvert$	$\lvert P^{\lambda_i}_{i_L(n+1)}\rvert$	$\lvert P^{\lambda_i}_{v_C(n+1)}\rvert$	$\lvert P^{\lambda_i}_{\varphi(n+1)}\rvert$
λ_1	0.0000	0.7491	0.0003	0.6196	0.5304
λ_2	0.0000	0.7491	0.0003	0.6196	0.5304
λ_3	0.0000	0.0000	0.9993	0.0007	0.0000
λ_4	0.0000	0.7511	0.0000	0.0110	0.2599

the battery R_{eq}, and the proportional constant k_p of the controller have the significant influence on the system stability. Therefore, according to the proposed stability criteria of different types of instability phenomena, the parameter space of these system parameters can be obtained. The detailed parameter spaces are shown in Figure 19. Because the model proposed in this paper is relatively accurate and the criteria of all possible instability phenomena are considered, the parameter design and stability boundary proposed in this paper are very accurate, which can provide the guidance for the practical parameters design.

3.2 | CP charging mode stability assessment

Due to the process of stability analysis is the same as CC charging mode, the results of CP charging mode are introduced directly below.

3.2.1 | Correlation factor

The correlation factors between the state variables and Floquet multipliers of CP charging mode are shown in Table 4 and Figure 20. It is obvious that the fixed Floquet multiplier λ_4 is closely related with the capacitor voltage v_C. In addition, the inductor current i_L is closely related to the Floquet multiplier λ_3 which is the same as CC charging mode. Besides, the phase shift angle φ and the capacitor voltage v_C is closely related with the Floquet multiplier λ_1 and λ_2, which could result in the low-frequency oscillation.

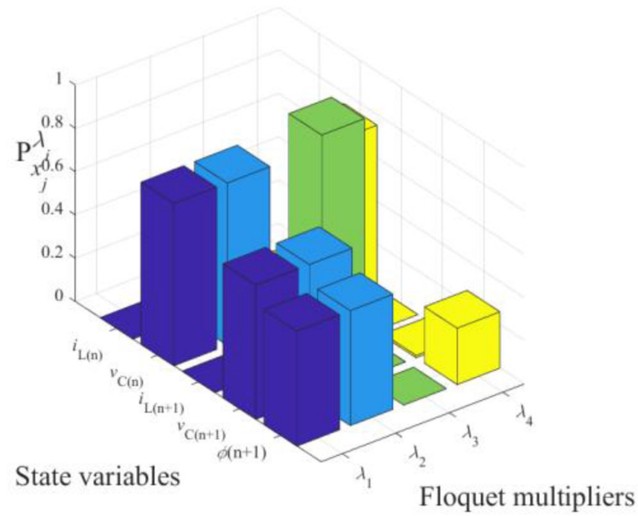

FIGURE 20 The CP charging mode correlation factors between the state variables and Floquet multipliers

3.2.2 | Sensitivity of the Floquet multipliers

Table 5 shows the parameter sensitivity of the Floquet multipliers of the DAB converter. The sensitivity of the Floquet multipliers is illustrated in Figure 21. Similar to CC charging mode, the conjugate pair of Floquet multipliers λ_1, λ_2 have very low sensitivity to system parameters like V_2, R_t, and the output power reference P_ref. Floquet multipliers λ_1, λ_2 are related with the V_1, C, L, R_{eq} and k_p. Moreover, the trend of system instability caused by the change of these parameters is the same as that of CC charging mode which is not discussed due to the paper length limitation.

3.2.3 | Parameter spaces and stability boundaries

Figure 22 shows the parameter spaces of the DC bus voltage V_1, the equivalent series resistance of the battery R_{eq}, and the proportional constant k_p.

TABLE 5 Parameter sensitivity of the DAB converter

	λ_1	λ_2	λ_3	λ_4
$S_{V_1}^{\lambda_i}$	$0.0011 + 0.4135i$	$0.0011 - 0.4135i$	$-0.0006 + 0.0000i$	$-0.0018 + 0.0000i$
$S_{V_2}^{\lambda_i}$	$-0.0002 + 0.0197i$	$-0.0002 - 0.0197i$	$0.0005 + 0.0000i$	$-0.0001 + 0.0000i$
$S_{P_{ref}}^{\lambda_i}$	$0.0000 + 0.0124i$	$0.0000 - 0.0124i$	0.0000+0.0000	$-0.0001 + 0.0000i$
$S_{R_{eq}}^{\lambda_i}$	$0.0777 - 0.4213i$	$0.0777 + 0.4213i$	$0.0084 + 0.0000i$	$0.0000 + 0.0000i$
$S_{R_t}^{\lambda_i}$	$-0.0046 + 0.0162i$	$-0.0046 - 0.0162i$	$-0.1618 + 0.0000i$	$0.0000 + 0.0000i$
$S_{L}^{\lambda_i}$	$0.0037 - 0.4503i$	$0.0037 + 0.4503i$	$0.1618 + 0.0000i$	$0.0019 + 0.0000i$
$S_{C}^{\lambda_i}$	$0.0777 - 0.4324i$	$0.0777 + 0.4324i$	$0.0085 + 0.0000i$	$-0.0001 + 0.0000i$
$S_{k_p}^{\lambda_i}$	$0.0009 + 0.4356i$	$0.0009 - 0.4356i$	$0.0000 + 0.0000i$	$-0.0020 + 0.0000i$
$S_{k_I}^{\lambda_i}$	$0.0037 + 0.0006i$	$0.0037 - 0.0006i$	$0.0000 + 0.0000i$	$-0.0074 + 0.0000i$

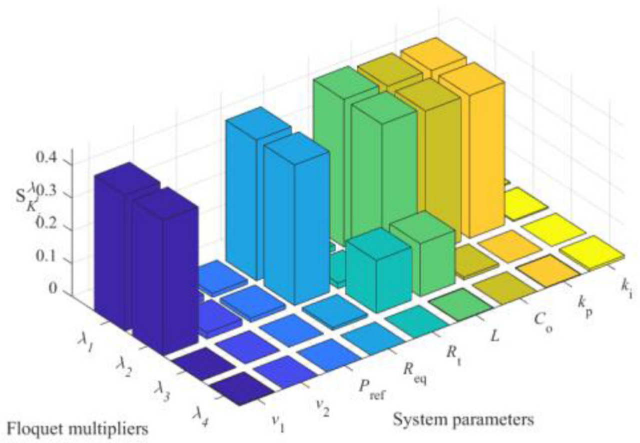

FIGURE 21 The CP charging mode sensitivity of the Floquet multipliers

3.3 | CV charging mode stability assessment

3.3.1 | Correlation factor

The correlation factors between the state variables and Floquet multipliers of CV charging mode are shown in Table 6 and Figure 23. Unlike CC charging mode and CP charging mode, the fixed Floquet multiplier λ_4 is a little related with the capacitor voltage v_C. In addition, the inductor current i_L is closely related to the Floquet multiplier λ_3 which is the same as CC charging mode. Besides, the phase shift angle φ and the capacitor voltage v_C is related with the Floquet multiplier λ_1 and λ_2, which could result in the low-frequency oscillation.

TABLE 6 Correlation factor $|\mathbf{P}_{ij}|$

	$\lvert P_{i_L(n)}^{\lambda_i}\rvert$	$\lvert P_{v_C(n)}^{\lambda_i}\rvert$	$\lvert P_{i_L(n+1)}^{\lambda_i}\rvert$	$\lvert P_{v_C(n+1)}^{\lambda_i}\rvert$	$\lvert P_{\varphi(n+1)}^{\lambda_i}\rvert$
λ_1	0.0000	1.7946	0.0011	1.1238	0.8225
λ_2	0.0000	1.7946	0.0011	1.1238	0.8225
λ_3	0.0000	0.0001	0.9982	0.0018	0.0001
λ_4	0.0000	0.5553	0.0000	0.0142	0.4588

3.3.2 | Sensitivity of the Floquet multipliers

Table 7 shows the parameter sensitivity of the Floquet multipliers of the DAB converter. The sensitivity of the Floquet multipliers is illustrated in Figure 24. The conjugate pair of

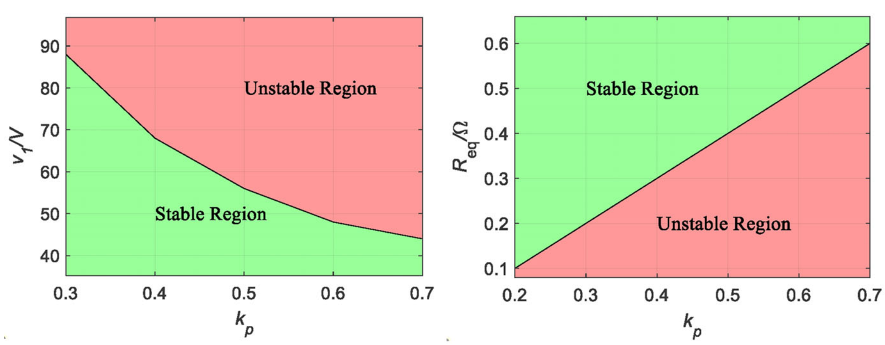

FIGURE 22 The parameter spaces and the stability boundaries of CP charging mode

TABLE 7 Parameter sensitivity of the DAB converter

	λ_1	λ_2	λ_3	λ_4
$S_{V_1}^{\lambda_i}$	$0.0016 + 0.4850i$	$0.0016 - 0.4850i$	$-0.0006 + 0.0000i$	$-0.0024 + 0.0000i$
$S_{V_2}^{\lambda_i}$	$-0.0001 + 0.0246i$	$-0.0001 - 0.0246i$	$0.0005 + 0.0000i$	$-0.0001 + 0.0000i$
$S_{v_{ref}}^{\lambda_i}$	$0.0000 + 0.0000i$	$0.0000 + 0.0000i$	$0.0000 + 0.0000i$	$0.0000 + 0.0000i$
$S_{R_{eq}}^{\lambda_i}$	$0.0785 - 0.0900i$	$0.0785 + 0.0900i$	$0.0092 + 0.0000i$	$-0.0025 + 0.0000i$
$S_{R_t}^{\lambda_i}$	$-0.0051 + 0.0228i$	$-0.0051 - 0.0228i$	$-0.1618 + 0.0000i$	$0.0001 + 0.0000i$
$S_{L}^{\lambda_i}$	$0.0036 - 0.5345i$	$0.0036 + 0.5345i$	$0.1608 + 0.0000i$	$0.0024 + 0.0000i$
$S_{C}^{\lambda_i}$	$0.0771 - 0.6022i$	$0.0771 + 0.6022i$	$0.0093 + 0.0000i$	$-0.0000 + 0.0000i$
$S_{k_p}^{\lambda_i}$	$0.0011 + 0.5129i$	$0.0011 - 0.5129i$	$0.0000 + 0.0000i$	$-0.0024 + 0.0000i$
$S_{k_I}^{\lambda_i}$	$0.0027 - 0.0026i$	$0.0027 + 0.0026i$	$0.0000 + 0.0000i$	$-0.0055 + 0.0000i$

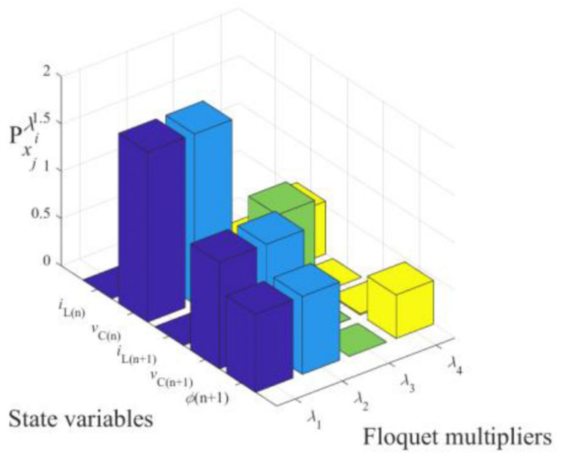

FIGURE 23 The CV charging mode correlation factors between the state variables and Floquet multipliers

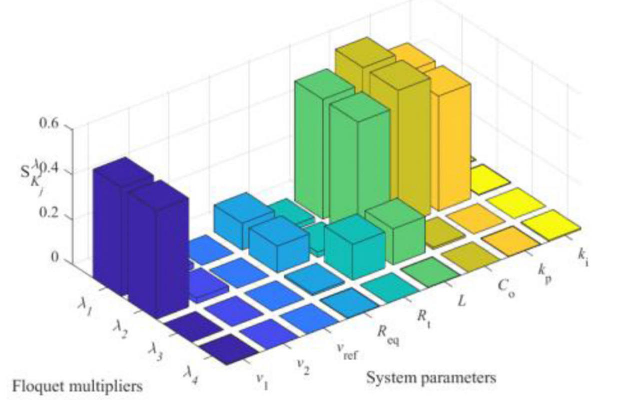

FIGURE 24 The CV charging mode sensitivity of the Floquet multipliers

Floquet multipliers λ_1, λ_2 have low sensitivity to system parameters like V_2, R_t, R_{eq}, and the output voltage reference v_{ref}. Floquet multipliers λ_1, λ_2 are related with the V_1, C, L, and k_{p}.

3.3.3 | Parameter spaces and stability boundaries

Figure 25 shows the parameter spaces of the DC bus voltage V_1, the equivalent series resistance of the battery R_{eq}, and the proportional constant k_{p}.

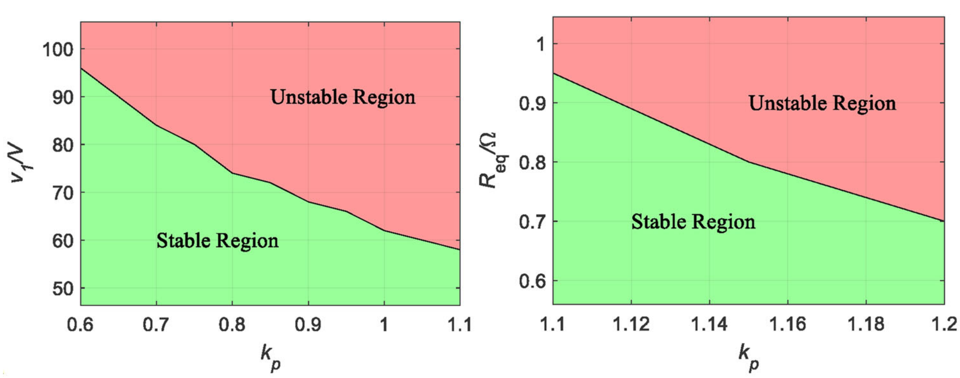

FIGURE 25 The parameter spaces and the stability boundaries of CV charging mode

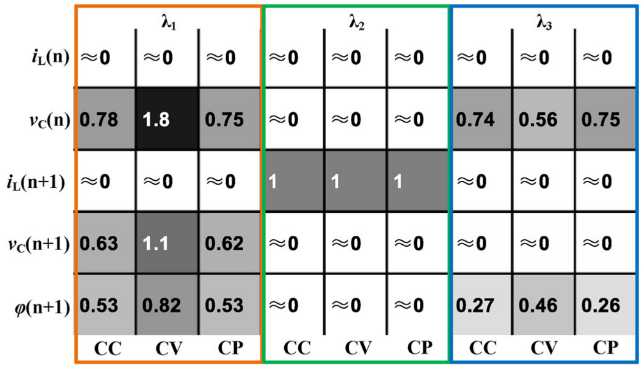

FIGURE 26 Comparison of correlation factor of three charging modes

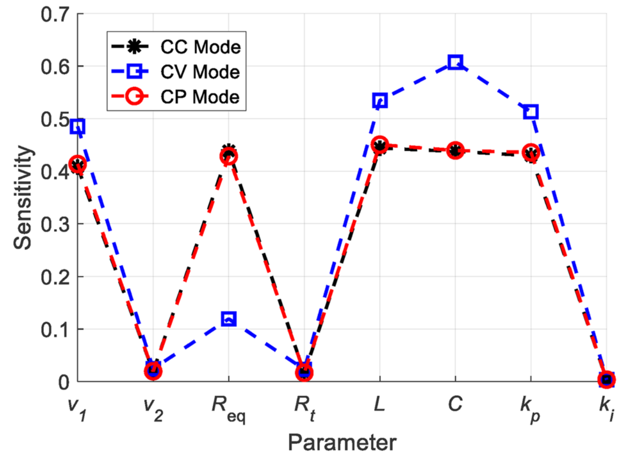

FIGURE 27 Comparison of sensitivity of three charging modes

4 | DISCUSSIONS AND PARAMETER DESIGN

4.1 | Discussions

4.1.1 | Comparison of correlation factor for three charging modes

Figure 26 shows the comparison of correlation for three charging modes (CC, CP, and CV). Due to the conjugate pair of Floquet multipliers λ_1, λ_2 have the same correlation factor, Figure 26 only consider λ_1. The number in the figure represents the value of the corresponding correlation factor, and the colour of each colour block represents the value of the number. The darker the colour, the higher the correlation between state variables and the Floquet multipliers. For λ_1, it can be seen that the correlation factor of CC charging mode is similar to CP charging mode. For CV charging mode, the phase shift angle φ and the capacitor voltage v_C is closely related with the Floquet multiplier $\lambda 1$, which could result in the low-frequency oscillation.

4.1.2 | Comparison of sensitivity for three charging modes

Figure 27 shows the comparison of sensitivity for three charging modes (CC, CP, and CV). It can be seen that the sensitivity of CC mode and CP mode are consistent which is because their goal is to control the output current in fact. The stability of three charging modes is not associated with the battery voltage V_2, equivalent series resistance R_t, integral constant k_I and the output reference value.

The stability of three charging modes is related to the inductance L, DC bus voltage V_1, output capacitor C, equivalent series resistance of the battery R_{eq}, and the digital the proportional constant k_p. Furthermore, the CV charging mode is more relevant to V_1, L, C, and k_p. However, CV charging mode is not related with R_{eq} when comparing with CC and CP charging mode.

4.2 | Stability-oriented parameter design of the system

The stability of the system for three charging modes are all related with L, V_1, C, R_{eq}, and k_p. Through the above analysis, the following conclusions can be drawn when designing the parameters of the system.

1. For capacitor, it is noticeable that the signs of the image parts of $S_C^{\lambda_1}$ and $S_C^{\lambda_2}$ are opposite to those of λ_1 and λ_2, which indicates that λ_1 and λ_2 will gradually move away from the unit circle with the increase of C and the system stability margin will accordingly expand. But in the actual selection, the size and cost of the capacitor must be considered to determine the appropriate capacitor value for the DAB converter. In addition, as time goes by, the capacitance value will gradually decrease, and the system will be at risk of instability. Therefore, after using the capacitor for a long time, it is recommended to check the capacitance of the capacitor.
2. The increase of L will expand the stability margin. Therefore, when designing the inductor of DAB converter, it is essential to ensure the inductance value is not too small.
3. With the number of battery charging and discharging increases, the internal resistance of the battery R_{eq} will increase. But the system will become more stable. This shows that the stability of the system based on three charging modes has little to do with the age of the battery.
4. For control parameters, the system has high sensitivity to the proportional constant k_p and low sensitivity to the integral constant k_I. The system becomes unstable as k_p increases. However, it is well-known that the dynamic response will be faster with larger k_p. Therefore, when designing the control parameters, trial and error should be taken according to practical requirements.
5. The oscillation of the DC-bus voltage V_1 may cause the instability of the system. Compared with CC and CP charging modes, the CV charging mode has larger stability boundaries of V_1. Therefore, it is important for the DAB

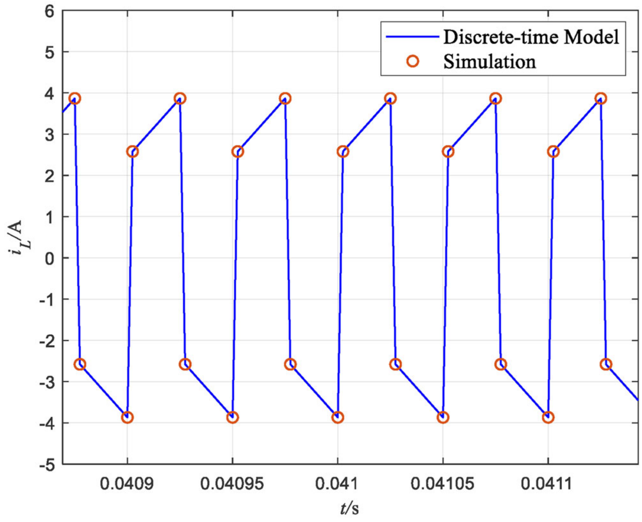

FIGURE 28 Inductor current waveform at steady state

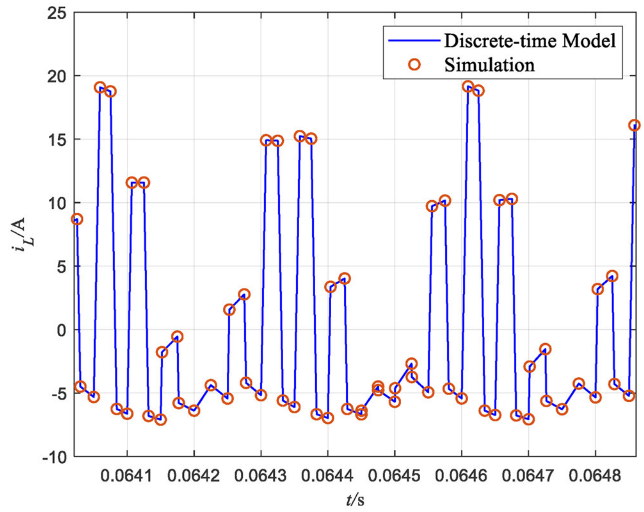

FIGURE 29 Inductor current waveform of unstable state

converter working in CC and CP mode to design appropriate k_p when there is fluctuation of V_1.

6. The system has very low sensitivity to system parameters like battery voltage V_2, equivalent series resistance R_t, the output reference value. This shows that when the external parameters of the system undergo sudden changes, such as battery voltage and reference value step-change etc., the system can still be stable.

5 | SIMULATION AND EXPERIMENTAL VERIFICATION

Simulations and experiments are carried out in this part to validate the theoretical analysis. The objective of the simulation experiments is to show that the discrete-time model provides the same accuracy as the simulation. The objective of the hardware validation is to verify the stability region analysis.

5.1 | Simulation results

One of the key applications of the proposed discrete-time model for three charging modes is efficient and accurate time-domain simulation. To verify the accuracy of the model, results of system-level simulations performed in MATLAB/SIMULINK are compared with results obtained through iteration of the model. System parameters are listed in Table 1.

Comparison of simulation and discrete-time model of CC charging mode at steady state is shown in Figure 28. Comparisons of simulation and discrete-time model of CC charging mode at unstable state is shown in Figure 29. In the process of using the discrete-time model to draw the waveform, we record the current value at the beginning of the four state subintervals in one switching cycle. The blue solid line in the figure is drawn by the model while the red circle is the current value obtained by simulation. It can be seen from Figures 28 and 29 that the proposed model can accurately track the simulation waveform of the system with high accuracy. The states not shown in these figures share the similar accuracy.

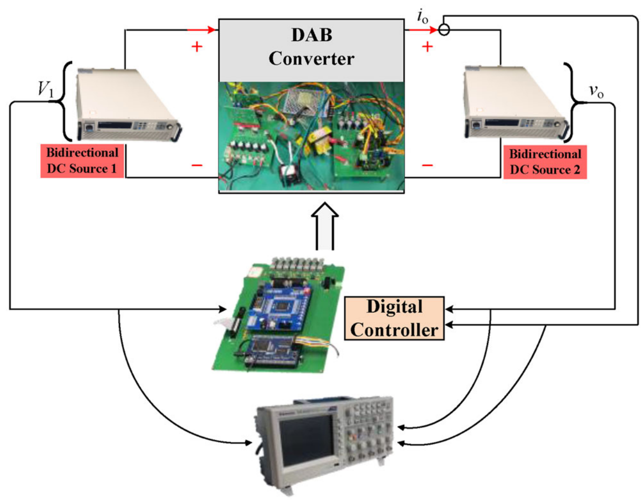

FIGURE 30 Schematic diagram of the experimental platform

5.2 | Experimental verification

Furthermore, in order to verify the theoretical analysis and simulation results, an experimental prototype with the same design parameter values used in simulation is built as shown in Figures 30 and 31. In the experiment, the two sides of DAB converter are the bidirectional DC sources respectively. Furthermore, the bidirectional DC source can be regarded as an energy storage battery by setting the internal resistance. The input and output voltages and output current signals will be transmitted to the ADC module of the TMS320F28335 DSP controller through the sensing and conditioning circuit. Then, the digital controller calculates the phase shift angle of the next switching

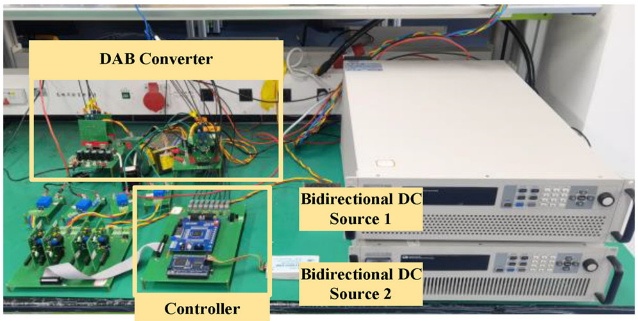

FIGURE 31 The experimental platform

cycle according to the different charging mode. Then through the drive circuits, the signals are sent to the DAB converter to achieve the control goal.

For CC charging mode, the inductor current i_L, the output current i_o, and the input voltage V_1 waveforms of the DAB converter system when $V_1 = 48$ V are shown in Figure 32a. When the proportional constant k_p is 0.63, it can be observed that the converter is stable and there is no DC component in the inductor current i_L. However, when k_p increases to 0.8, the system is unstable and the inductor current will exhibit the slow-scale low-frequency oscillation as shown in Figure 32b. When the proportional constant k_p is 0.63, the system is unstable when $V_1 = 53$ V as shown in Figure 32c. The experimental results are corresponding to the parameter space shown in Figure 19.

For CP charging mode, the inductor current i_L, the output current i_o, and the input voltage V_1 waveforms of the DAB converter system when $V_1 = 48$ V are shown in Figure 33a. When the proportional constant k_p is 0.58, it can be observed that the converter is stable and there is no DC component in the inductor current i_L. However, when k_p increases to 0.7, the system is unstable and the inductor current will exhibit the slow-scale low-frequency oscillation as shown in Figure 33b. When the proportional constant k_p is 0.58, the system is unstable when $V_1 = 53$ V as shown in Figure 33c. The experimental results are corresponding to the parameter space shown in Figure 22.

For CV charging mode, the inductor current i_L, the output current i_o, the input voltage V_1, and the output voltage v_o waveforms of the DAB converter system when $V_1 = 48$ V are shown in Figure 34a. When the proportional constant k_p is 1.3, it can be observed that the converter is stable and there is no DC component in the inductor current i_L. However, when k_p increases to 1.5, the system is unstable and the inductor current will exhibit the slow-scale low-frequency oscillation as shown in Figure 34b. When the proportional constant k_p is 1.3, the system is unstable when $V_1 = 53$ V as shown in Figure 34c.

The experimental waveforms clearly depict the same dynamical behaviour as the analytical and simulation results, and it is obvious that there are slow-scale low-frequency oscillations on the waveforms of the inductor current when k_p and V_1 are greater than the critical value. Furthermore, it is proved that the inductor current under unstable condition has the characteristics of high amplitude and asymmetry. The asymmetrical inductor current will cause DC bias and the flux bias in the transformer which will eventually lead to the transformer saturation and the sharp increase in current. It could increase the equipment stress or destruct the power switches [14]. The oscillation of the output current also decreases the quality of the power supply. These phenomena are not expected and can be eliminated by reasonable design parameters. In addition, to verify the theoretical analysis and experimental results, the Bode diagrams for three charging modes are obtained as shown in Figure 35, which are corresponding to the experimental results in Figures 32–34. It is clear that when the system is stable, the phase margin in the Bode diagram is positive. However, when the system is unstable which means the slow-scale low-frequency oscillation occurs, the phase margin is negative. The Bode diagrams prove the correctness of the experimental results.

Due to the influence of core loss, various parasitic elements, dead effect, the amplitude of experimental waveform is not exactly the same as the simulation result, and there exists small deviations of k_p between experimental results and theoretical predictions. Nevertheless, these small errors do not affect the verification of the experiment.

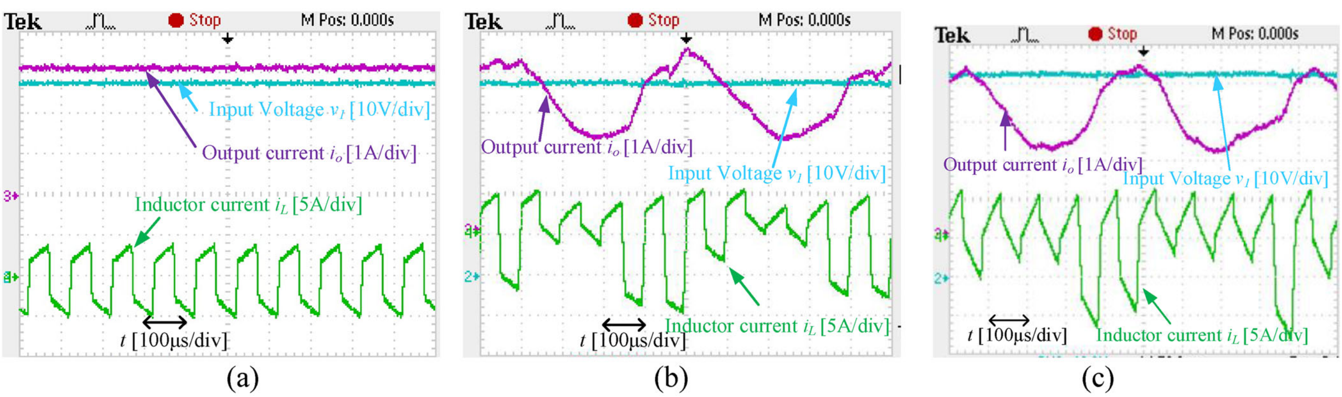

FIGURE 32 The inductor current, input voltage and output current of the DAB converter system (CC mode). (a) $V_1 = 48$ V $k_p = 0.63$, (b) $V_1 = 48$ V $k_p = 0.8$, (c) $V_1 = 53$ V $k_p = 0.63$

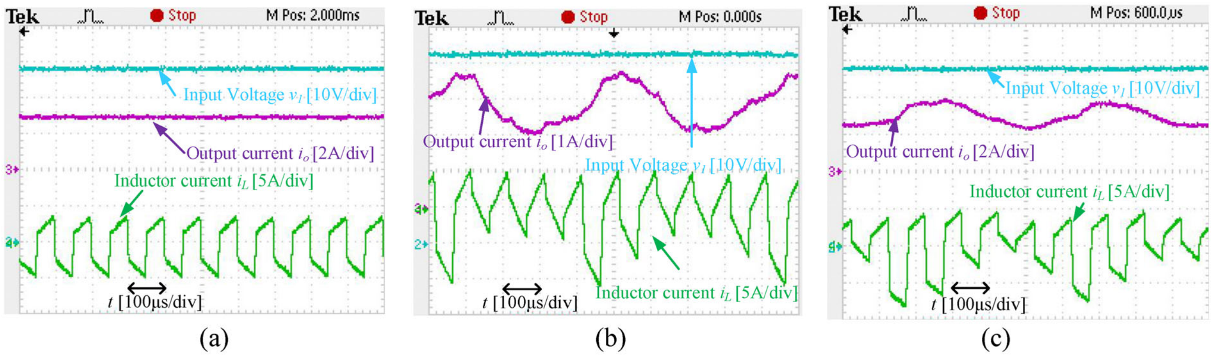

FIGURE 33 The inductor current, input voltage and output current of the DAB converter system (CP mode). (a) $V_1 = 48$ V $k_p = 0.58$, (b) $V_1 = 48$ V $k_p = 0.7$, (c) $V_1 = 53$ V $k_p = 0.58$

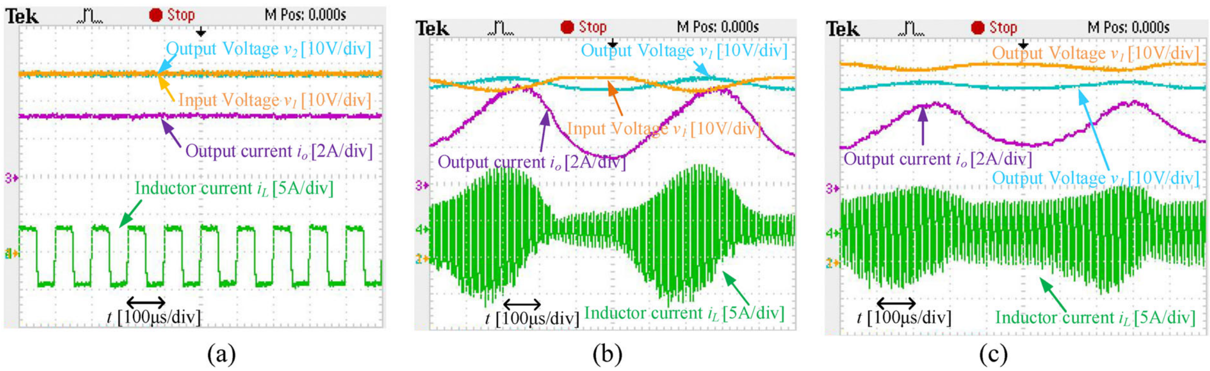

FIGURE 34 The inductor current, input voltage, output voltage, and output current of the DAB converter system (CV mode). (a) $V_1 = 48$ V $k_p = 1.3$, (b) $V_1 = 48$ V $k_p = 1.5$, (c) $V_1 = 53$ V, $k_p = 1.3$

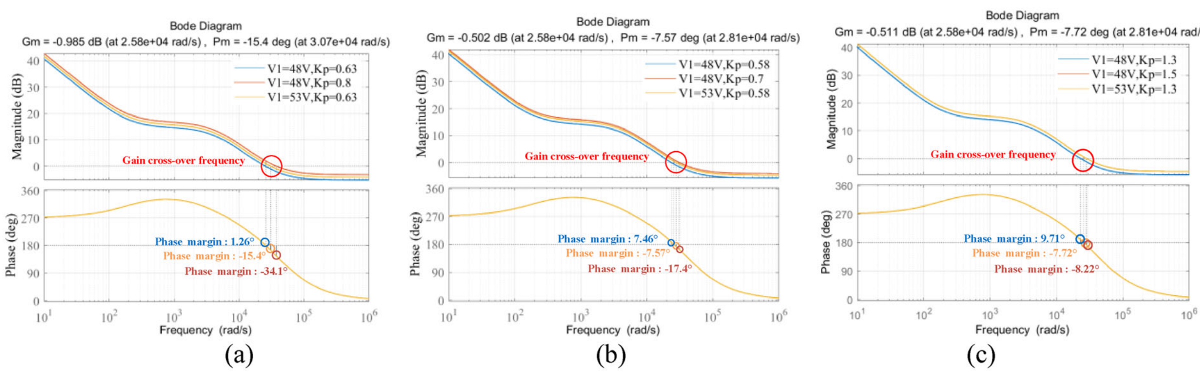

FIGURE 35 The Bode diagrams for three charging modes corresponding the experimental results. (a) CC, (b) CP, (c) CV

6 | CONCLUSION

The stability of DAB converter integrated with ESS in DC microgrid has been systematically analyzed under all charging modes (CC, CP, CV). Based on the accurate discrete-time model approach, correlation factor, and sensitivity analysis, the relationships among the state variables, Floquet multipliers and system parameters are revealed. It is demonstrated that the inductor L, DC bus voltage V_1, output capacitor C, equivalent series resistance of the battery R_{eq}, and the proportional constant k_{p} can influence the stability of the system greatly especially for CC charging mode and CP charging mode. The system will be more stable with the increase of L and C of the DAB converter and equivalent series resistance of the battery R_{eq}. The stability of the system based on three charging modes has little to do with the age of the battery. Meanwhile, the parameter spaces of the significant system parameters are obtained. By comparing the stability boundaries among different charging modes, the CV charging mode has larger stability range when there is the oscillation of the dc-bus voltage V_1. Therefore, it is essential for the DAB converter working in CC and CP mode to design appropriate k_{p} under the fluctuation of

V_1. Conclusions in this paper can provide the reliable guidance for the parameters design and ensure the stable operation of the system. Finally, the theoretical analysis under three operating modes is verified by simulations and experimental results. We believe that the models and the stability analysis of three charging modes for ESS proposed in this paper are useful in the field of energy storage charging in DC microgrid.

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CONFLICT OF INTEREST

No

DATA AVAILABILITY STATEMENT

Research data are not shared.

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